

Assuming that there is a supermassive black hole at the center of Sgr A* and that it is actually at the center of our Galaxy's gravitational well, then it should remain nearly motionless. Studies that have attempted to measure the proper motion of Sgr A* with respect to extremely distant extragalactic sources known as quasars (see Chapter 28) indicate that to within the uncertainties in the measurements, Sgr A* reflects only the peculiar motion of the Sun, just what would be expected for an object at rest at the center of the Galaxy. Furthermore, relative to other objects in the Galactic center, Sgr A* has moved at most a small fraction of its diameter over a period of more than a decade, a rate that is much slower than for any other object in the region. Given the gravitational tugs that it certainly receives from stars and large clouds in that region, this extremely sluggish motion implies that Sgr A* is very massive.

Very high-resolution VLA maps at 2 cm reveal what appear to be large clouds of ionized gas only 0.06 pc from Sgr A*. From the orientation of the clouds, it appears that they were ejected from the center in opposite directions. There is also evidence of a trail of hot, ionized gas coming from IRS 7, a red supergiant star less than 0.3 pc from the center. The direction of the stream points away from Sgr A*, suggesting that a strong wind or perhaps intense UV radiation from the center is blowing the material away from the star.

As has already been mentioned, although the Galactic center appears relatively quiet today, it must have experienced rather violent episodes in the recent past. One possible mechanism that could produce these periodic events would be the tidal disruption of a passing star and the subsequent infall of matter. As the matter falls toward the supermassive black hole in the center, it collects onto an accretion disk, releasing a tremendous amount of gravitational potential energy accompanied by a dramatic increase in luminosity (see Section 18.2). The entire episode may last for only a few years, but if such events occurred on the order of every 10^4 to 10^5 years, enough energy would be released to keep the central cavity ionized and the circumnuclear disk turbulent.

Sgr A* is not entirely inactive, however. From studies of the Galactic center using the Chandra X-Ray Observatory and the XMM-Newton Observatory, it appears that flares occur on average about once per day. Lasting for up to an hour or so, luminosities can reach peak values of 3.6×10^{28} W, more than 160 times the quiescent X-ray level of Sgr A*.

Many other galaxies (Andromeda, for example) appear to have supermassive black holes in their centers and yet are relatively quiet (see Section 25.2). On the other hand, other galaxies have extremely active nuclei. These active galactic nuclei probably derive their energies from central “engines” that are powered by supermassive black holes. Active galactic nuclei will be discussed in more detail in Chapter 28.

High-Energy Emission Lines from Near the Galactic Center

Before leaving the discussion of the Galactic center, it is worth mentioning the detection of two significant high-energy emission lines that originate in the region of the Sgr A complex. The first of these results from an electron colliding with a positron, causing their mutual annihilation and the corresponding production of two photons, each of energy 511 keV.¹² Since it is believed that black holes can help produce positrons in the space surrounding

¹²511 keV is the rest mass energy of an electron or a positron.

them, the presence of the 511-keV line seemed to support the existence of a black hole in Sgr A*. However, the efficient production of positrons in the numbers needed to explain the enormous flux of 10^{44} photons per second ($L_{511} \sim 5 \times 10^4 L_{\odot}$) requires a black hole smaller than the one envisioned for the Galactic center (perhaps only a few hundred solar masses). This is because high accretion-disk temperatures are required for the production of the particles, and the temperature of the disk increases with decreasing radius (see Eq. 18.19).

Due to the poor angular resolution of the original detections, it was unclear whether Sgr A* could properly be considered the source of the 511-keV photons. On October 13–14, 1990, the Soviet spacecraft GRANAT, with its higher-resolution imaging capability, discovered that the source of the 511-keV photons was not Sgr A* but a previously known X-ray emitter, 1E1740.7–2942, first detected by the Einstein satellite in 1979.¹³ Nicknamed “the Einstein source,” 1E1740.7–2942 is located some 45' from Sgr A*, more than 300 pc from the center. With its accretion-disk plasma temperature of 10^9 K and variable luminosity, the Einstein source appears to be a very strong candidate for a stellar black hole.

The second high-energy emission line detected in the region of the Galactic center is the 1.8-MeV line produced by the decay of $^{26}_{13}\text{Al}$ to $^{26}_{12}\text{Mg}$. Since $^{26}_{13}\text{Al}$ has a half-life of 716,000 years and is produced only in relatively small amounts in supernovae, novae, and possibly Wolf–Rayet stars, the presence of an estimated $\sim 5 M_{\odot}$ of the radioactive isotope seems to suggest that a large number of supernovae have occurred in that region over the past 10^5 to 10^6 years. The Galactic center is clearly an extremely dynamic environment.

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¹³The source gets its name from its right ascension and declination.

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PROBLEMS

- 24.1** Approximately how many times has the Sun circled the center of the Galaxy since the star’s formation?
- 24.2** (a) What fraction of the total *B*-band luminosity of the Galaxy is produced by each of the stellar components? Refer to Table 24.1.
(b) What is the total bolometric magnitude of the Galaxy?
- 24.3** The globular cluster IAU C0923–545 has an integrated apparent visual magnitude of $V = +13.0$ and an integrated absolute visual magnitude of $M_V = -4.15$. It is located 9.0 kpc from Earth and is 11.9 kpc from the Galactic center, just 0.5 kpc south of the Galactic midplane.
(a) Estimate the amount of interstellar extinction between IAU C0923–545 and Earth.
(b) What is the amount of interstellar extinction per kiloparsec?
- 24.4** Using the differential star count formula for an infinite universe of constant stellar number density and no interstellar extinction (Eq. 24.5), show that the amount of light arriving at Earth from a cone of solid angle Ω diverges exponentially as the length of the cone increases without bound (or, equivalently, as m approaches infinity). Assume that all stars in the field have the same absolute magnitude M . *Hint:* You may find the discussion leading up to Eq. (3.5) helpful.

TABLE 24.2 Hypothetical Differential Star Count Data.

V	$\log_{10} A_M$	V	$\log_{10} A_M$
4	-2.31	12	2.24
5	-1.71	13	2.59
6	-1.11	14	2.94
7	-0.51	15	3.29
8	0.09	16	3.89
9	0.69	17	4.49
10	1.29	18	5.09
11	1.89	19	5.69

- 24.5 (a)** From Eq. (24.5), derive an expression for $\log_{10} A_M(m)$ as a function of m for stars of the same absolute magnitude and M–K spectral classification, assuming a constant stellar number density.
- (b)** If observations are made in apparent magnitude bins separated by one (i.e., $\delta m = 1$), calculate

$$\Delta \log_{10} A_M(m) \equiv \log_{10} A_M(m+1) - \log_{10} A_M(m).$$

- (c)** If the results of observations show that $\Delta \log_{10} A_M(m)$ is always less than the result found in part (b), what can you conclude about the distribution of stars in the region under investigation? [Recall that Eq. (24.5) applies to the case of an infinite universe of constant stellar number density and no interstellar extinction.]
- 24.6 (a)** Plot $\log_{10} A_M$ as a function of V for the hypothetical data given in Table 24.2. Assume that all stars included in the differential star counts are main-sequence A stars of absolute visual magnitude $M_V = 2$.
- (b)** Assuming a constant density of stars out to at least $V = 11$, how much interstellar extinction is present to that limit (express your answer in magnitudes)? *Hint:* Consider the slope of the curve. You may also find the results of Problem 24.5 helpful.
- (c)** What is the distance to the stars corresponding to $V = 11$?
- (d)** If the solid angle over which the data were collected is 0.75 square degrees, or 2.3×10^{-4} sr, estimate the number density $n_M(M, S)$ of A stars out to $V = 11$.
- (e)** Give two possible explanations for the change in slope between $V = 11$ and $V = 15$.
- 24.7 (a)** Assume that a cloud of gas and dust is encountered along the line of sight for the data given in Table 24.2 and plotted in Problem 24.6. Assume also that the stellar number density found in Problem 24.6 is constant along the entire line of sight. Estimate the amount of extinction (in magnitudes) that is due to the cloud. *Hint:* How would the graph change if the cloud were not present? The cloud's presence may be revealed through reddening.
- (b)** If the density of gas and dust in the cloud leads to an extinction rate of 10 mag kpc^{-1} , what is the length of the cloud along the line of sight?
- Graphs of $\log_{10} A_M$ vs. m that demonstrate changes in slope and then resume the original slope at larger values of m are referred to Wolf diagrams, after Maximilian Wolf (1863–1932), who first used them to explore the properties of interstellar clouds.
- 24.8 (a)** Plot the thin disk's luminosity density (Eq. 24.10) as a function of z for $R = 8 \text{ kpc}$.

- (b) Prove that for $z \gg z_0$,

$$L(R, z) \simeq 4L_0 e^{-R/h_R} e^{-2z/z_0}$$

and so $z_0 = 2z_{\text{thin}}$ is the effective scale height of the luminosity density function.

- 24.9** (a) From the data given in Table 24.1, and, using a typical value for the temperature of hydrogen in the interstellar medium of 15 K, estimate the average thermal energy density of hydrogen gas in the disk of the Galaxy. For this problem, assume that the disk has a radius of 8 kpc and a height of 160 pc.
- (b) Using Eq. (11.9), estimate the energy density of the magnetic field in the spiral arms. Compare your answer with the thermal energy density of the gas. Would you expect the magnetic field to play a significant role in the structure of the Galaxy? Why or why not?
- 24.10** What are the J2000.0 Galactic coordinates of Sgr A*?
- 24.11** Use Eqs. (24.16–24.18) to determine the Galactic coordinates of the following objects. (You may wish to refer to Fig. 24.18 to verify your answers.)
- (a) The north celestial pole
- (b) The vernal equinox
- (c) Deneb (see Appendix E)
- 24.12** (a) Estimate the height (z) above or below the Galactic plane for both M13 ($\ell = 59.0^\circ$, $b = 40.9^\circ$) and the Orion nebula ($\ell = 209.0^\circ$, $b = -19.4^\circ$). M13 and the Orion Nebula are 7 kpc and 450 pc from Earth, respectively.
- (b) To which components of the Galaxy do these objects probably belong? Explain your answers.
- 24.13** (a) Consider a sample of stars that lie in the Galactic plane and are distributed in a circle about the LSR, as shown in Fig. 24.36. For the purpose of this problem, assume also that these stars are at rest with respect to the LSR (of course, this could not actually occur in such a

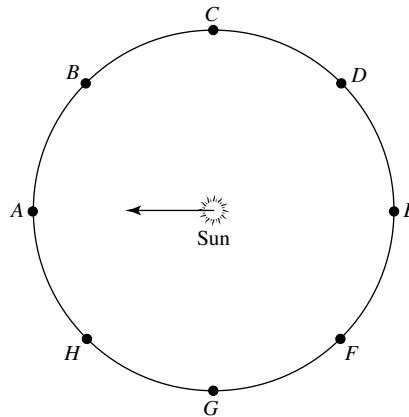


FIGURE 24.36 A set of stars distributed in a circle about the LSR. The circle is assumed to be in the Galactic plane, and the stars are at rest with respect to the LSR. The solar motion is in the direction of Star A.

dynamic system). With the Sun located at the position of the LSR and the solar motion in the direction of Star A as indicated, sketch the velocity vectors associated with the apparent motion of each star, as seen from the Sun. Label the apex and antapex on your diagram.

- (b) Sketch the radial-velocity and transverse-velocity components of each star's apparent motion on the diagram used in part (a).
- (c) Describe how you might locate the apex of the solar motion given the radial-velocity data of a large sample of stars in the solar neighborhood.
- (d) How would you identify the solar apex from proper motion data of stars in the solar neighborhood?

24.14 Figure 24.37 illustrates older data derived from a kinematic study of the Milky Way. From the data presented, what was the estimate of $v_{\odot}$ in the early 1960s?

- 24.15** (a) Assuming (incorrectly) that the high-velocity stars known to Oort in 1927 are near the escape speed from the Galaxy, estimate the mass of the Milky Way. For simplicity, take the directions of the velocity vectors to be radially away from the Galactic center and assume that all of the mass is spherically distributed and is interior to R_0 . (This calculation is meant only to be an order-of-magnitude estimate.) Compare your answer with the mass estimate given in Example 24.3.1.
- (b) Repeat your calculation using the extremely high-velocity stars discussed on page 908. What could account for the extra mass compared to your answer in part (a)?
- (c) Comment on the difficulty of determining the true mass of the Galaxy on the basis of observations of stars in the solar neighborhood.

24.16 Starting with Eqs. (24.37) and (24.38), derive Eqs. (24.41) and (24.42), showing each step explicitly.

24.17 Referring to Eq. (24.42) and Fig. 24.23, explain the functional dependence of transverse velocity on Galactic longitude for stars near the Sun.

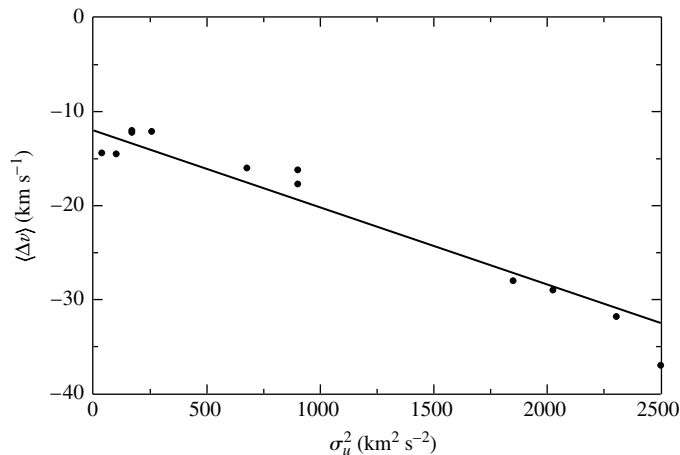


FIGURE 24.37 Each point represents a different sample of objects, including for instance supergiants, carbon stars, white dwarfs, Cepheids, and planetary nebulae. [Data from Delhaye, *Galactic Structure*, Blaauw and Schmidt (eds.), University of Chicago Press, Chicago, 1965.]

- 24.18** (a) Beginning with Kepler's third law (Eq. 2.37), derive an expression for $\Theta(R)$, assuming that the Sun travels in a Keplerian orbit about the center of the Galaxy.
 (b) From your result in part (a), derive analytic expressions for the Oort constants A and B .
 (c) Determine numerical values for A and B in the solar neighborhood, assuming $R_0 = 8$ kpc and $\Theta_0 = 220 \text{ km s}^{-1}$. Express your answers in units of $\text{km s}^{-1} \text{ kpc}^{-1}$.
 (d) Do your answers in part (c) agree with the measured values for the Milky Way Galaxy? Why or why not?
- 24.19** (a) Estimate $d\Theta/dR$ in the solar neighborhood, assuming that the Oort constants A and B are $+14.8$ and $-12.4 \text{ km s}^{-1} \text{ kpc}$, respectively. What does this say about the variation of Θ with R in the region near the Sun?
 (b) If A and B were $+13$ and $-13 \text{ km s}^{-1} \text{ kpc}$, respectively, what would the value of $d\Theta/dR$ be? What would this say about the shape of the rotation curve in the solar neighborhood?
- 24.20** (a) Show that rigid-body rotation near the Galactic center is consistent with a spherically symmetric mass distribution of constant density.
 (b) Is the distribution of mass in the dark matter halo (Eq. 24.51) consistent with rigid-body rotation near the Galactic center? Why or why not?
- 24.21** Using the result of the "back-of-the-envelope" calculation for the density of dark matter (Eq. 24.50), estimate the mass density of dark matter in the solar neighborhood. Express your answer in units of kg m^{-3} , $M_\odot \text{ pc}^{-3}$, and $M_\odot \text{ AU}^{-3}$. How does your answer compare with the stellar mass density in the solar neighborhood?
- 24.22** (a) Assuming that Eq. (24.51) is valid for any arbitrary distance from the center of the Galaxy, show that the amount of dark matter interior to a radius r is given by the expression
- $$M_r = 4\pi\rho_0 a^2 \left[r - a \tan^{-1} \left(\frac{r}{a} \right) \right].$$
- (b) If $5.4 \times 10^{11} M_\odot$ of dark matter is located within 50 kpc of the Galactic center, determine ρ_0 in units of $M_\odot \text{ pc}^{-3}$. Repeat your calculation if $1.9 \times 10^{12} M_\odot$ is located within 230 kpc of the Galactic center. Assume that $a = 2.8$ kpc.
- 24.23** Using Eq. (24.52) for the density profile of the dark matter halo, show that
 (a) $\rho_{\text{NFW}} \propto r^{-1}$ for $r \ll a$ and $\rho_{\text{NFW}} \propto r^{-3}$ for $r \gg a$.
 (b) the integral of the mass from $r = 0$ to $r \rightarrow \infty$ is infinite.
- 24.24** Using data provided in the text for the mass of the dark matter halo interior to 50 kpc and interior to 230 kpc, estimate the values for the constants ρ_0 and a in the NFW version of the dark matter halo density profile (Eq. 24.52).
- 24.25** (a) From the information given in Table 24.1 and in the text, determine the approximate mass-to-light ratio of the Galaxy interior to a radius of 25 kpc from the center.
 (b) Repeat your calculation for a radius of 100 kpc. What can you conclude about the effect that dark matter might have on the average mass-to-light ratio of the universe?
- 24.26** The r^{-2} dependence of Coulomb's electrostatic force law allows the construction of Gauss's law for electric fields, which has the form

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{in}}}{\epsilon_0},$$

where the integral is taken over a closed surface that bounds the enclosed charge, Q_{in} . Because Newton's gravitational force law also varies as r^{-2} , it is possible to derive a gravitational "Gauss's law." The form of this gravitational version is

$$\oint \mathbf{g} \cdot d\mathbf{A} = -4\pi G M_{\text{in}}, \quad (24.56)$$

where the integral is over a closed surface that bounds the mass M_{in} , and $\mathbf{g}$ is the local acceleration of gravity at the position of $d\mathbf{A}$. The differential area vector ($d\mathbf{A}$) is assumed to be normal to the surface everywhere and is directed outward, away from the enclosed volume.

Show that if a spherical gravitational Gaussian surface is employed that is centered on and surrounds a spherically symmetric mass distribution, Eq. (24.56) can be used to solve for $\mathbf{g}$. The result is the usual gravitational acceleration vector around a spherically symmetric mass.

- 24.27** We learned in Sections 24.2 and 24.3 that the Sun is currently located 30 pc north of the Galactic midplane and moving away from it with a velocity $w_{\odot} = 7.2 \text{ km s}^{-1}$. The z component of the gravitational acceleration vector is directed toward the midplane, so the Sun's peculiar velocity in the z direction must be decreasing. Eventually the direction of motion will reverse and the Sun will pass through the midplane heading in the opposite direction. At that time the direction of the z component of the gravitational acceleration vector will also reverse, ultimately causing the Sun to move northward again. This oscillatory behavior above and below the midplane has a well-defined period and amplitude that we will estimate in this problem.

Assume that the disk of the Milky Way has a radius that is much larger than its thickness. In this case, as long as we confine ourselves to regions near the midplane, the disk appears to be infinite in the $z = 0$ plane. Consequently, the gravitational acceleration vector is always oriented in the $\pm z$ direction. We will neglect the radial acceleration component in this problem.

- (a) By constructing an appropriate Gaussian surface and using Eq. (24.56), derive an expression for the gravitational acceleration vector at a height z above the midplane, assuming that the Sun always remains inside the disk of constant density ρ .
- (b) Using Newton's second law, show that the motion of the Sun in the z direction can be described by a differential equation of the form

$$\frac{d^2 z}{dt^2} + kz = 0.$$

Express k in terms of ρ and G . This is just the familiar equation for simple harmonic motion.

- (c) Find general expressions for z and w as functions of time.
- (d) If the total mass density in the solar neighborhood (including stars, gas, dust, and dark matter) is $0.15 M_{\odot} \text{ pc}^{-3}$, estimate the oscillation period.
- (e) By combining the current determinations of $z_{\odot}$ and $w_{\odot}$, estimate the amplitude of the solar oscillation and compare your answer with the vertical scale height of the thin disk.
- (f) Approximately how many vertical oscillations does the Sun execute during one orbital period around the Galactic center?

- 24.28** Show that

$$d = \frac{\langle v_r \rangle \tan \phi}{\langle \mu \rangle},$$

leads to Eq. (24.54) with the appropriate change in units.

24.29 Refer to the data in Appendix E for this problem.

- (a) What angle does Altair's space motion vector make with its radial-velocity vector?
- (b) Find Altair's transverse velocity and space motion relative to the Sun.

24.30 Using Newtonian gravity, estimate the amount of energy required to move $10^7 M_\odot$ from a position just above the event horizon of the supermassive black hole at the center of the Galaxy to 3 kpc, the present location of the expanding arm. Compare your answer to the amount of energy liberated in a typical Type II supernova.

24.31 If the accretion rate at the Galactic center is $10^{-3} M_\odot \text{ yr}^{-1}$ and if it has remained constant over the past 5 billion years, how much mass has fallen into the center over that period of time? Compare your answer with the estimated mass of a possible supermassive black hole residing in the center of our Galaxy.

24.32 (a) Compute the lowest possible density of Sgr A* based on the data obtained from the orbit of S2. Assume a spherically symmetric mass distribution.

- (b) Assuming a mass of $3.7 \times 10^6 M_\odot$ and a radius of 1 AU (roughly the current limit of resolution of the center of the Milky Way), estimate the density of Sgr A*. Express your answer in kg m^{-3} , $M_\odot \text{ AU}^{-3}$, and $M_\odot \text{ pc}^{-3}$.

24.33 Using the data found in the text, calculate the speed of S2 when it is closest to Sgr A*.

24.34 Using Newtonian gravity, estimate the Roche limit of a supermassive black hole of mass $3.7 \times 10^6 M_\odot$ (assume that a $1 M_\odot$ main-sequence star is tidally disrupted). How does your answer compare with the black hole's Schwarzschild radius? *Hint:* Begin with Eq. (19.4), substituting the appropriate average densities and radii.

24.35 Estimate the Eddington luminosity of a black hole with the mass of Sgr A*. What is the ratio of the upper limit of the bolometric luminosity of Sgr A* to its Eddington luminosity?

24.36 In this problem you will construct a crude model for the mass distribution and velocity curve in the inner 1 kpc of the Galaxy. Assume that a point (a black hole) of mass $M_0 = 3.7 \times 10^6 M_\odot$ is located at the center of the Galaxy and that the remainder of the mass has an isothermal density distribution that varies as r^{-2} .

- (a) Show that if the mass distribution is spherically symmetric, the mass interior to a radius r can be expressed as a function of the form

$$M_r = kr + M_0,$$

where k is a constant to be determined.

- (b) Assuming perfectly circular motion and Newtonian gravity, show that the orbital velocity curve is given by

$$v = \left[G \left(k + \frac{M_0}{r} \right) \right]^{1/2}.$$

- (c) If the orbital velocity is 110 km s^{-1} at 2 pc, determine a value for k .
- (d) Plot $\log_{10} M_r$ as a function of $\log_{10} r$ over the range $0.01 \text{ pc} < r < 1 \text{ kpc}$. Express M_r in solar units and r in parsecs. Your graph should be qualitatively similar to the observational data depicted in Fig. 24.33.
- (e) Plot v as a function of $\log_{10} r$ over the range $0.01 \text{ pc} < r < 1 \text{ kpc}$. Express v in km s^{-1} and r in parsecs. At what radius does the contribution of the central point mass begin to become significant?

CHAPTER

25

The Nature of Galaxies

- 25.1 *The Hubble Sequence*
- 25.2 *Spiral and Irregular Galaxies*
- 25.3 *Spiral Structure*
- 25.4 *Elliptical Galaxies*

25.1 ■ THE HUBBLE SEQUENCE

As was mentioned at the beginning of Chapter 24, it was in the middle of the eighteenth century that Kant and Wright first suggested that the Milky Way represents a finite-sized disk-like system of stars. In the two centuries of scientific investigation since their proposal, we have indeed come to learn that a major component of our Galaxy is well represented by a disk of stars that also contains a significant amount of gas and dust. As an extension of their philosophical argument about the nature of the Galaxy, Kant went on to suggest that if the Milky Way is limited in extent, perhaps the diffuse and very faint “elliptical nebulae” seen in the night sky might actually be extremely distant disk-like systems, similar to our own but well beyond its boundary.¹ He called these objects *island universes*.

Cataloging the Island Universes

The true nature of the island universes became a matter of much investigation, and extensive catalogs of these objects were collected. One such catalog we owe to Charles Messier (1730–1817), who, while hunting for comets, recorded 103 fuzzy objects that could otherwise be confused with the intended targets of his search.² Although many of the members of the **Messier catalog** are truly gaseous nebulae contained within the Milky Way (such as the Crab supernova remnant and the Orion Nebula, M1 and M42, respectively), and others are stellar clusters (for instance, the Pleiades open cluster is M45 and the great globular cluster in Hercules is M13), the nature of other nebulae, such as M31 in Andromeda (Fig. 24.7), was unknown.

Another catalog of nebulae was produced by William Herschel and subsequently expanded by his son, Sir John Herschel (1792–1871), to include the southern hemisphere.

¹Although today we generally use the word *nebula* to refer to gas and dust clouds, it was originally used to describe any fuzzy patch of light in the sky that could not be resolved into a clear ensemble of stars.

²Later, other astronomers added seven additional members to the original list, bringing the total number of Messier objects to 110. The Messier catalog is found in Appendix H.

Later, J. L. E. Dreyer (1852–1926) published the **New General Catalog** (NGC), which was based on the work of the Herschels and contained almost 8000 objects. Like Messier’s catalog, the NGC includes many entries that are either gaseous nebulae or stellar clusters located within the Milky Way. However, the true nature of other objects in the catalog remained in question.³

It was in 1845 that William Parsons, the third Earl of Rosse (1800–1867), built what was then the largest telescope in the world. Located in Ireland and nicknamed the “Leviathan,” the 72-in (1.8-m) instrument was able to resolve, for the first time, the spiral structure in some nebulae. Their pinwheel appearance strongly suggested that these *spiral nebulae* may be rotating. This suspicion was eventually verified by Vesto M. Slipher (1875–1969) in 1912 when he detected Doppler-shifted spectral lines in a number of these objects.

The Great Shapley–Curtis Debate

The argument over the nature of the nebulae centered on their distances from us and the relative size of the Galaxy. Many astronomers believed that the spiral nebulae resided within the confines of the Milky Way, and others favored the view that they were really Kant’s island universes. On April 26, 1920, at the National Academy of Sciences in Washington, D.C., Harlow Shapley of the Mount Wilson Observatory and Heber D. Curtis (1872–1932) of the Lick Observatory met to argue the merits of each point of view. In what has become known as the *Great Debate* in astronomy, Shapley supported the idea that the nebulae are members of our Galaxy. Curtis, on the other hand, was a proponent of the extragalactic interpretation of the data, believing that the nebulae were physically much like the Milky Way, but separated from it.

One of Shapley’s strongest points was based on the apparent magnitudes of novae observed in M31. He argued that if the disk of Andromeda were as large as the Milky Way (approximately 100 kpc in diameter by his own recent estimates), then its angular size in the sky ($\sim 3^\circ \times 1^\circ$) would imply a distance to the nebula that was so large as to make the luminosities of the novae in M31 much greater than those found in the Milky Way.

His second major point was based on data of Adrian van Maanen (1884–1946), a well-respected observer, whose proper-motion measurements of M101 seemed to suggest an angular rotation rate of $0.02'' \text{ yr}^{-1}$. If M101 had a diameter similar to Shapley’s estimate for the Milky Way, then points near its outer edge would have rotational speeds far in excess of those observed within the Milky Way.

In defense of the extragalactic hypothesis, Curtis argued that the novae observed in spiral nebulae must be at least 150 kpc away from us in order to have intrinsic brightnesses comparable to those in the Milky Way. At this distance, M31 would be similar in size to Kapteyn’s much smaller estimate of the diameter of the Galaxy, rather than to Shapley’s estimate (recall the discussion in Section 24.1). He also argued that the large radial velocities measured for many spiral nebulae seemed to indicate that they could not remain gravitationally bound within a Kapteyn-model Milky Way. Furthermore, assuming that the transverse velocities of the nebulae are similar in value to their radial velocities, then if the nebulae were close enough to be located within the Milky Way it should be possible to

³It is worth pointing out that most of the members in the Messier catalog are contained within the NGC; for instance, M31 is also designated as NGC 224.

measure their proper motions across the sky; however, no such motions had been detected. Finally, for those spiral nebulae that are oriented edge-on, dark absorption regions can be seen. Curtis suggested that if the Milky Way had a similar dark layer, the zone of avoidance would be easily explained.

In the end, neither set of arguments proved to be definitive and the Great Debate served more to highlight the issues than to solve the dilemma. Although, as we now know, errors existed on both sides of the controversy, Shapley's arguments were perhaps the more flawed. Part of the problem rested in his overestimate of the size of the Milky Way's disk. The other difficulty was in his reliance on van Maanen's data, which van Maanen himself later showed to be incorrect. In fact, proper-motion studies were unable to measure any rotation of M101.

The debate was finally settled in 1923 when Edwin Hubble (1889–1953) detected Cepheid variable stars in M31 using the 100-inch telescope at Mount Wilson. By measuring their apparent magnitudes and determining their absolute magnitudes via the period–luminosity relation (Eq. 14.1), he was able to use the distance modulus $m - M$ (Eq. 3.5) to calculate the distance to Andromeda. Hubble's original value of 285 kpc is approximately 2.7 times smaller than the modern estimate of 770 kpc, but it was still good enough to show definitively that the spiral nebulae are indeed island universes.

The process of scientific exploration that led to the realization that our Solar System is not located near the center of the Milky Way, and that the Galaxy is only one of a countless number of galaxies in the universe, represented an advance in our understanding of the universe comparable to that of the Copernican Revolution. Although many people contributed to that new understanding, it was in large part Hubble's pivotal work in determining extragalactic distances that set the stage for many subsequent investigations of galaxies, the largest stellar systems in the universe.⁴

The Classification of Galaxies

Now that the extragalactic nature of the galaxies had been established, work began on determining their physical properties. As a first step in understanding any new collection of objects, it is necessary to classify them according to their intrinsic characteristics, akin to the zoological classification of various species of animals. Once again Hubble played a key role. In his 1926 paper “Extra-Galactic Nebulae,” and later in his book *The Realm of the Nebulae*, Hubble proposed that galaxies be grouped into three primary categories based on their overall appearance. This morphological classification scheme, known as the **Hubble sequence**, divides galaxies into **ellipticals** (E's), **spirals**, and **irregulars** (Irr's).⁵ The spirals are further subdivided into two parallel sequences, the **normal spirals** (S's), and the **barred spirals** (SB's). A transitional class of galaxies between ellipticals and spirals, known as **lenticulars**, can be either normal (SO's) or barred (SBO's). Hubble then arranged his morphological sequence in the form of a **tuning-fork diagram**, shown in Fig. 25.1, which explicitly shows the two types of spirals. A galaxy's **Hubble type** is its designation along the Hubble sequence.

⁴The challenging and important process of determining the extragalactic distance scale, to which Hubble made another critical contribution, is the subject of Sections 27.1 and 27.2.

⁵A more modern classification designates true irregulars as Ir, as shown in Fig. 25.1.

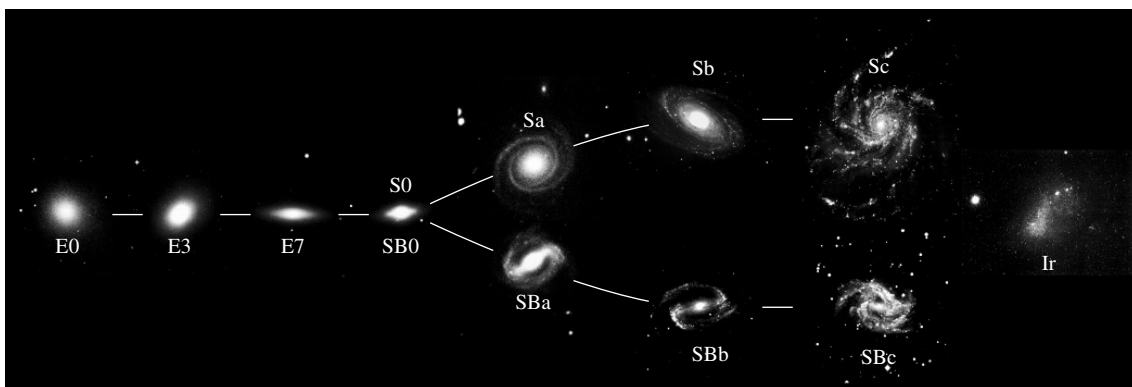


FIGURE 25.1 Hubble's tuning-fork diagram of galaxy types.

Hubble originally thought (incorrectly) that the tuning-fork diagram could be interpreted as an evolutionary sequence for galaxies. As a result, he referred to galaxies toward the left of the diagram as *early types* and to those toward the right as *late types*, terminology that is still in widespread use today.

Within the category of ellipticals, Hubble made divisions based on the observed **ellipticity** of the galaxy, defined by

$$\epsilon \equiv 1 - \beta/\alpha, \quad (25.1)$$

where α and β are the *apparent* major and minor axes of the ellipse, respectively, projected onto the plane of the sky. The Hubble type is then quoted in terms of 10ϵ . Ellipticals range from a spherical distribution of stars, E0, to a highly flattened distribution, E7. Galaxies with ellipticities greater than $\epsilon = 0.7$ have never been observed, implying that no E galaxies with intrinsic ellipticities greater than 0.7 appear to exist.

It is important to realize that a difficulty immediately arises with this scheme: The apparent ellipticity may not correspond well to an actual ellipticity since the orientation of the spheroid⁶ to our line of sight plays a crucial role in our observations. This effect can be seen in Figs. 25.2 and 25.3, when observers view **oblate** and **prolate** galaxies, respectively. Note that, in general, $a \geq b \geq c$ represents the lengths of the three axes of a **triaxial** spheroidal system. For a sphere $a = b = c$, for a perfectly oblate spheroid $a = b$, and for a perfectly prolate spheroid $b = c$. In general, however, there is no requirement that any of the axes of a spheroidal system have equal lengths.

The physical properties of elliptical galaxies cover an enormous range. Their absolute B magnitudes may be as dim as -8 or brighter than -23 , their masses (including both luminous and dark matter) vary from as little as $10^7 M_\odot$ to more than $10^{13} M_\odot$, and their diameters can be as small as a few tenths of a kiloparsec or as large as hundreds of kiloparsecs. The giant elliptical galaxies are among the largest objects in the universe, while the smallest dwarfs are comparable in size to a typical globular cluster. The lenticular galaxies have masses and luminosities comparable to the larger ellipticals. Although the giant ellipticals and the

⁶The term *spheroid* is commonly used to mean *axisymmetric ellipsoid*.

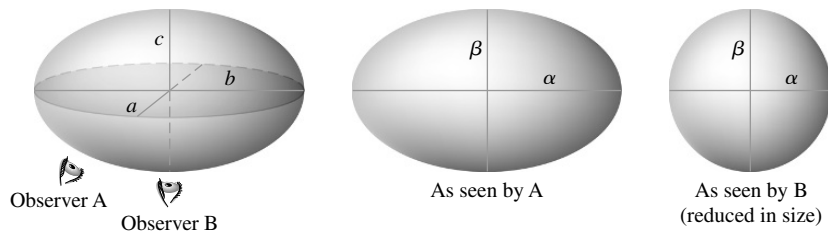


FIGURE 25.2 An oblate spheroidal galaxy has axis lengths $a = b$ and $c < a$. If $c/a = 0.6$, the apparent shape resembles an E4 galaxy ($\beta/\alpha = 0.6$) when seen by observer A. The same galaxy appears as an E0 when seen by observer B ($\beta/\alpha = 1$).

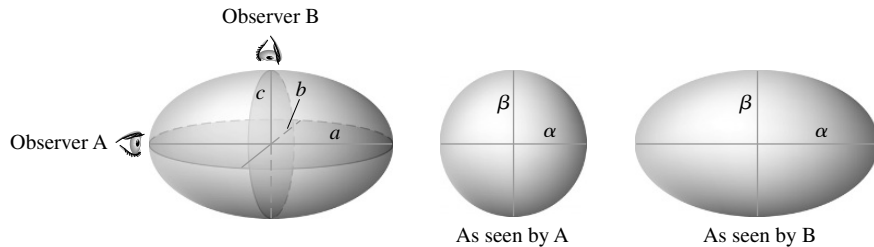


FIGURE 25.3 A prolate spheroidal galaxy has axis lengths $b = c$ and $a < b$. If $b/a = 0.6$, the apparent shape resembles an E0 galaxy when seen by observer A. The same galaxy appears as an E4 when seen by observer B.

lenticulars are the easiest to observe, the dwarfs are by far the most numerous. Typical examples of elliptical galaxies are shown in Fig. 25.4, along with an S0 and an SB0. The physical characteristics of these galaxies will be discussed in more detail in Section 25.4.

Hubble subdivided the spiral sequences into Sa, Sab, Sb, Sbc, Sc, and SBa, SBab, SBb, SBbc, SBc. The galaxies with the most prominent bulges (the largest bulge-to-disk luminosity ratios, $L_{\text{bulge}}/L_{\text{disk}} \sim 0.3$), the most tightly wound spiral arms (with pitch angles⁷ of approximately 6°), and the smoothest distribution of stars in the arms are classified as Sa's (or SBa's), while Sc's (or SBc's) have smaller bulge-to-disk ratios ($L_{\text{bulge}}/L_{\text{disk}} \sim 0.05$), more loosely wound spiral arms ($\sim 18^\circ$), and spiral arms that resolve into clumps of stars and H II regions. Examples of normal and barred spiral galaxies are shown in Figs. 25.5 and 25.6, respectively. M31 (see Fig. 24.7) and NGC 891 (see Fig. 24.9) are Sb's, whereas the Milky Way is probably an SBbc (Fig. 24.5 reflects the evidence for a central bar in our Galaxy).

Sa–Sc (SBa–SBc) galaxies tend to have much smaller variations in their physical parameters than do ellipticals. On average, spirals also tend to be among the largest galaxies in the universe, with absolute B magnitudes from -16 to less than -23 , masses (including both luminous and dark matter) between $10^9 M_\odot$ and $10^{12} M_\odot$, and disk diameters of 5 to 100 kpc.

⁷The *pitch angle* is defined as the angle between the tangent to the spiral arm and the tangent to a perfect circle, measured at the point where the arm and the circle intersect.

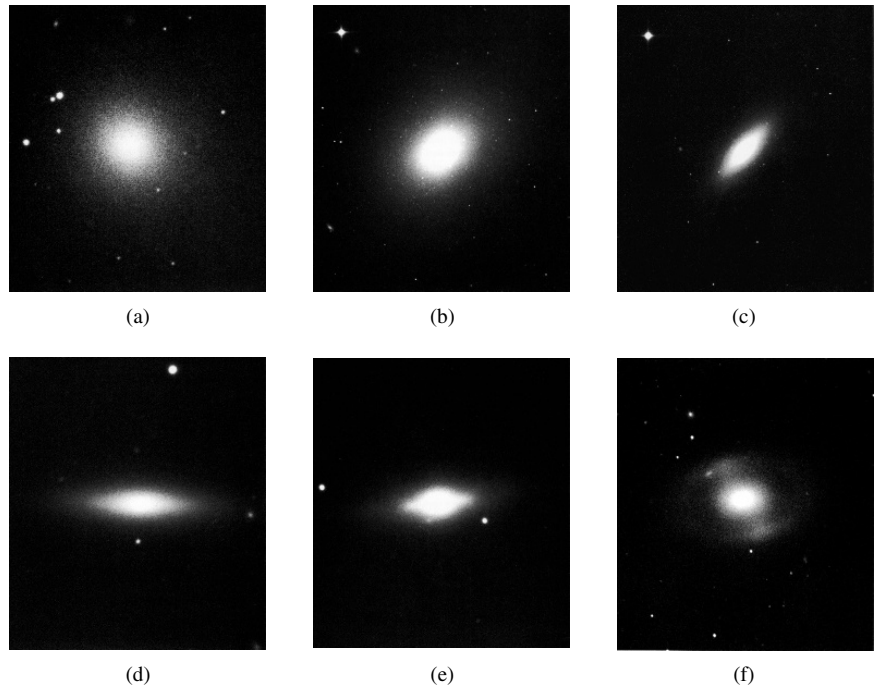


FIGURE 25.4 Typical early-type galaxies. (a) IC 4296 (E0), (b) NGC 4365 (E3), (c) NGC 4564 (E6), (d) NGC 4623 (E7), (e) NGC 4251 (S0), (f) NGC 4340 (RSB0). (Images from Sandage and Bedke, *The Carnegie Atlas of Galaxies*, Carnegie Institution of Washington, Washington, D.C., 1994.)

Hubble split the remaining category of irregulars into Irr I if there was at least some hint of an organized structure, such as spiral arms, and Irr II for the most extremely disorganized structures. Both the Large Magellanic Cloud and the Small Magellanic Cloud are examples of Irr I galaxies, while M82 (NGC 3034) is an example of an Irr II; see Fig. 25.7.

Irregular galaxies have a wide range of characteristics, although they tend not to be particularly large. Typically their absolute B magnitudes vary from -13 to -20 , they have masses of between $10^8 M_{\odot}$ and $10^{10} M_{\odot}$, and their diameters range from 1 to 10 kpc. Most irregulars also tend to have noticeable bars that are often off-center.

Since the publication of Hubble's tuning-fork diagram, astronomers have made numerous modifications to his original classification scheme. For instance, Gerard de Vaucouleurs suggested the elimination of the irregular classifications, Irr I and Irr II, in favor of the addition of other morphological classes later than Sc (or SBc). Those galaxies that were binned into Irr I have been designated Sd (SBd), Sm (SBm), or Im (where m stands for Magellanic type); for instance, the LMC is classified as an SBm, and the SMC is an Im. The truly irregular galaxies are simply designated Ir, such as M82 (which is why Ir is used as the designation in Fig. 25.1). Sandage and Brucato further suggested that the Ir class should more appropriately be termed **amorphous** to indicate the lack of any organized structure. As we shall see, spirals of Hubble-type Sd and later tend to be significantly smaller than earlier-type spirals; consequently, they are sometimes referred to as **dwarf spirals**.

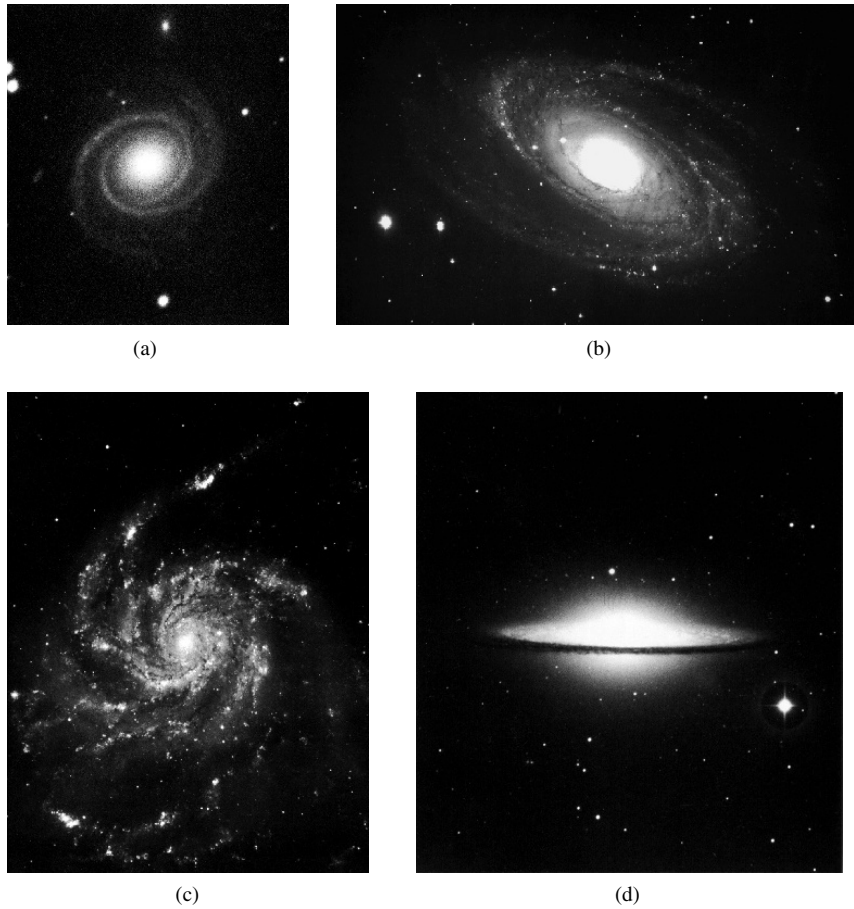


FIGURE 25.5 Typical normal spirals. (a) NGC 7096 (Sa(r)I), (b) M81/NGC 3031 (Sb(r)I–II), (c) M101/NGC 5457/Pinwheel (Sc(s)I), (d) M104/NGC 4594/Sombrero (Sa/Sb) seen nearly edge on. (Images from Sandage and Bedke, *The Carnegie Atlas of Galaxies*, Carnegie Institution of Washington, Washington, D.C., 1994.)

In order to make finer distinctions between normal and barred spirals, de Vaucouleurs had also suggested referring to normal spirals as SA rather than simply S. Intermediate types with weak bars are then characterized as SAB, and strongly barred galaxies are SB.

As a further refinement to the system, the lenticular galaxies are also sometimes subdivided according to the amount of dust absorption in their disks. S0₁ galaxies have no discernable dust in their disks, while S0₃ galaxies have significant amounts of dust, and similarly for SB0₁ through SB0₃.

Thus, the modern sequence from early ellipticals through normal late-type galaxies is (see also Fig. 25.1)

E0, E1, . . . , E7, S0₁, S0₂, S0₃, Sa, Sab, Sb, Sbc, Sc, Scd, Sd, Sm, Im, Ir.

A similar sequence exists for barred spirals.

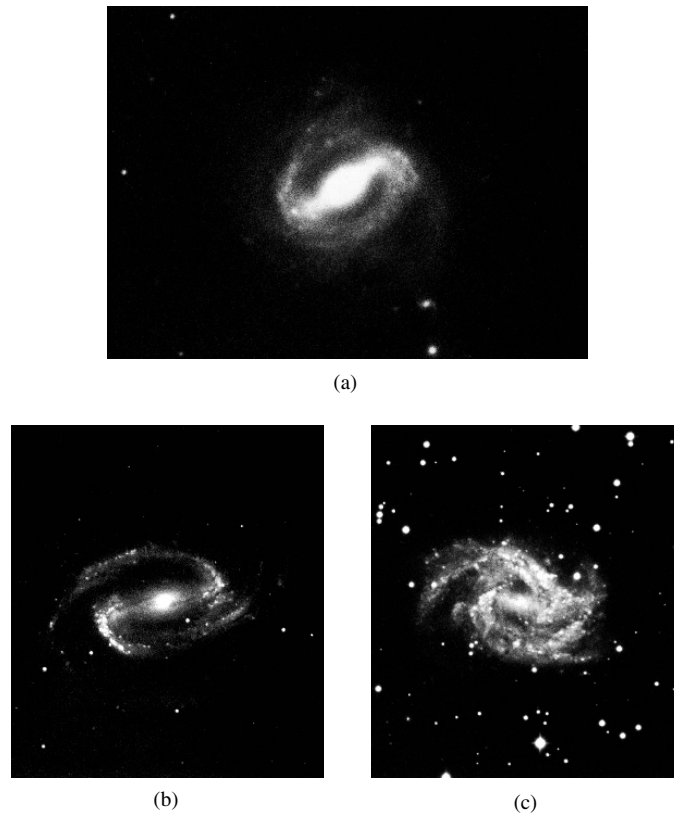


FIGURE 25.6 Typical barred spirals. (a) NGC 175 (SBab(s) I–II), (b) NGC 1300 (SBb(s) I), (c) NGC 2525 (SBc(s)II). (Images from Sandage and Bedke, *The Carnegie Atlas of Galaxies*, Carnegie Institution of Washington, Washington, D.C., 1994.)

Sidney van den Bergh introduced the **luminosity class** for spirals. The class ranges from I through V, with I representing those spirals with well-defined arms; galaxies with the least distinct arms are classified as V. M31 is classified as SbI–II (intermediate between I and II), the Milky Way Galaxy is an SBbcI–II, M101 is an ScI, the LMC is an SBmIII, and the SMC is an ImIV–V. Except for the largest elliptical galaxies (to be discussed further in Section 25.4), the Milky Way and M31 are among some of the largest and brightest galaxies in the universe. It is important to note, however, that despite its name, luminosity class does not necessarily correlate well with absolute magnitude.

Besides their striking arms, spirals also show an amazing array of more complex and subtle features. While some galaxies have spiral arms that can be followed nearly all the way into the center, others have arms that appear to terminate at the location of an inner ring. Special designations further help to classify these systems. M101 [Fig. 25.5(c)] is a galaxy of the former type and is labeled as an Sc(s)I, where (s) designates that the spiral can be traced to the center of the galaxy. On the other hand, NGC 7096 [Fig. 25.5(a)] and M81 [Fig. 25.5(b)] are galaxies of the later type and are classified as Sa(r)I and Sb(r)I–II,

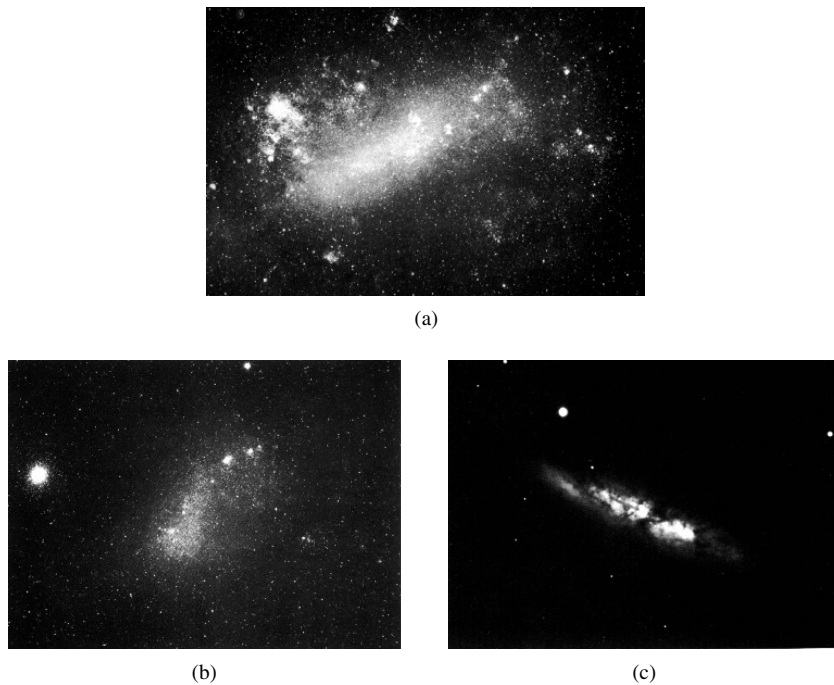


FIGURE 25.7 Examples of irregular galaxies. (a) The Large Magellanic Cloud (Irr I/SBmIII), (b) the Small Magellanic Cloud (Irr I/ImIV–V), (c) M82/NGC 3034 (Irr II/Ir/Amorphous). (Images from Sandage and Bedke, *The Carnegie Atlas of Galaxies*, Carnegie Institution of Washington, Washington, D.C., 1994.)

respectively, where (r) indicates an inner ring. Galaxies may also have outer rings that can be identified. One such example is NGC 4340 [Fig. 25.4(f)], which is designated as an RSB0, where the R prefix stands for outer ring. In some cases, galaxies may have both an inner ring and an outer ring.

Given the enormous range in galaxy morphological types, it is not surprising that any classification scheme will necessarily be complex. Fortunately, the Hubble sequence, with its various permutations and enhancements, has greatly facilitated the process of trying to understand the nature of galaxies.

25.2 ■ SPIRAL AND IRREGULAR GALAXIES

Hubble's classification scheme for late-type galaxies has proved to be very successful in organizing our study of these objects. Not only do bulge-to-disk ratios, the tightness of the spiral arms, and the ability to resolve the arms into stars and H II regions all correlate well with Hubble type, but so do a host of other physical parameters.

Tables 25.1 and 25.2 summarize the characteristics of late-type galaxies, the details of which will be discussed later. Although the spread in parameters can be quite large, trends in Hubble type are clearly evident. For instance, if we compare an Sa galaxy with an Sc

TABLE 25.1 Characteristics of Early Spiral Galaxies.

	Sa	Sb	Sc
M_B	−17 to −23	−17 to −23	−16 to −22
M ($M_\odot$)	10^9 – 10^{12}	10^9 – 10^{12}	10^9 – 10^{12}
$\langle L_{\text{bulge}}/L_{\text{total}} \rangle_B$	0.3	0.13	0.05
Diameter (D_{25} , kpc)	5–100	5–100	5–100
$\langle M/L_B \rangle$ ($M_\odot/L_\odot$)	6.2 ± 0.6	4.5 ± 0.4	2.6 ± 0.2
$\langle V_{\text{max}} \rangle$ (km s $^{-1}$)	299	222	175
V_{max} range (km s $^{-1}$)	163–367	144–330	99–304
pitch angle	$\sim 6^\circ$	$\sim 12^\circ$	$\sim 18^\circ$
$\langle B - V \rangle$	0.75	0.64	0.52
$\langle M_{\text{gas}}/M_{\text{total}} \rangle$	0.04	0.08	0.16
$\langle M_{\text{H}_2}/M_{\text{H I}} \rangle$	2.2 ± 0.6 (Sab)	1.8 ± 0.3	0.73 ± 0.13
$\langle S_N \rangle$	1.2 ± 0.2	1.2 ± 0.2	0.5 ± 0.2

TABLE 25.2 Characteristics of Late Spiral and Irregular Galaxies.

	Sd/Sm	Im/Ir
M_B	−15 to −20	−13 to −18
M ($M_\odot$)	10^8 – 10^{10}	10^8 – 10^{10}
Diameter (D_{25} , kpc)	0.5–50	0.5–50
$\langle M/L_B \rangle$ ($M_\odot/L_\odot$)	~ 1	~ 1
V_{max} range (km s $^{-1}$)	80–120	50–70
$\langle B - V \rangle$	0.47	0.37
$\langle M_{\text{gas}}/M_{\text{total}} \rangle$	0.25 (Scd)	0.5–0.9
$\langle M_{\text{H}_2}/M_{\text{H I}} \rangle$	0.03–0.3	~ 0
$\langle S_N \rangle$	0.5 ± 0.2	0.5 ± 0.2

galaxy of comparable luminosity, the Sa will be more massive (larger M/L_B), have a higher peak in its rotation curve (V_{max}), have a smaller mass fraction of gas and dust, and contain a higher proportion of older, red stars.

The K-Correction

Before we can consider these correlations in more detail, it is necessary to discuss the problems associated determining the brightnesses of galaxies. As is the case with determining absolute magnitudes for stars, calculating the absolute magnitudes of galaxies requires making corrections to their observed apparent magnitudes if we are to properly account for the effects of extinction, both within the Milky Way and within the target galaxy (see Eq. 12.1). (Extinction is generally negligible in the nearly empty space between galaxies.) Furthermore, for extragalactic objects another important correction must be considered as well.

Since most galaxies are observed to have measurable redshifts (the topic of Section 27.2), some or even most of the light that would normally fall within the wavelength band of interest, say the B band, would be redshifted to longer-wavelength regions. Accounting for this effect is known as the **K -correction**. As we will learn, the K -correction is most severe for very distant galaxies. If this effect were not considered, possible conclusions about the evolution of galaxies would be likely to be in error (see Sections 26.2 and 29.4).

The Brightness of the Background Sky

Another problem that arises when making observations of faint galaxies, or when measuring their outermost regions, is the competition with the brightness of the background sky. The dimly glowing night sky has an average surface brightness of about $\mu_{\text{sky}} = 22$ B -mag arcsec $^{-2}$ (measured in the B wavelength band). Sources of this background light include light pollution from nearby cities, photochemical reactions in Earth's upper atmosphere, the zodiacal light, unresolved stars in the Milky Way, and unresolved galaxies. However, in modern photometric studies using CCDs, the surface brightnesses of galaxies can be measured down to levels of 29 B -mag arcsec $^{-2}$ or fainter. Consequently, to accurately determine the light distribution of a galaxy at these extremely faint levels, it is necessary to subtract the contribution from the background sky.

Isophotes and the de Vaucouleurs Profile

Once the sky subtraction is performed, it becomes possible to map contours of constant surface brightness. Such contours are known as **isophotes** (lines of constant photon number). In specifying the “radius” of a galaxy, it is necessary to define the surface brightness of the isophote being used to determine that radius. We encountered this problem in trying to identify a radius for our own Galaxy in Chapter 24; no definite cutoff is known to exist in either the exponential distribution of the disk or the $r^{1/4}$ distribution of the spheroid. One commonly used radius, introduced by Erik Holmberg, a Swedish astronomer, is the **Holmberg radius**, r_H , defined to be the projected length of the semimajor axis of an ellipsoid having an isophotal surface brightness of $\mu_H = 26.5$ B -mag arcsec $^{-2}$. A second standard radius in frequent use is the **effective radius**, r_e , the projected radius within which one-half of the galaxy's light is emitted.⁸ The surface brightness level at r_e , designated μ_e , depends on the distribution of the surface brightness with radius.

For the bulges of spiral galaxies, and for large ellipticals, the surface brightness distribution typically follows an $r^{1/4}$ law given by

$$\mu(r) = \mu_e + 8.3268 \left[\left(\frac{r}{r_e} \right)^{1/4} - 1 \right]. \quad (25.2)$$

This is just the $r^{1/4}$ de Vaucouleurs profile discussed in Section 24.2 (Eq. 24.13), written in units of mag arcsec $^{-2}$, rather than $L_{\odot}$ pc $^{-2}$. It is left as an exercise (Problem 25.5) to show that these two forms are equivalent.

⁸The effective radius was first introduced on page 892.

Disks are frequently modeled with an exponential decay, similar to the one given in Eq. (24.10) for our Galaxy. However, as with the $r^{1/4}$ law, the disk luminosity per unit area can be written in units of mag arcsec^{-2} as

$$\mu(r) = \mu_0 + 1.09 \left(\frac{r}{h_r} \right), \quad (25.3)$$

where h_r is the characteristic scale length of the disk along its midplane (see Problem 25.6).

A generalized version of the $r^{1/4}$ law is frequently used in which $1/4$ is replaced by $1/n$. The resulting **generalized de Vaucouleurs profile**, also known as the **Sérsic profile** (named for José Luis Sérsic), has the form

$$\mu(r) = \mu_e + 8.3268 \left[\left(\frac{r}{r_e} \right)^{1/n} - 1 \right], \quad (25.4)$$

where μ_e , r_e , and n are all free parameters used to obtain the best possible fit to the actual surface brightness profile. Note that when $n = 1$, Eq. (25.3) is a special case of Eq. (25.4), where μ_0 and h_r are written in terms of μ_e and r_e , respectively.

The Rotation Curves of Galaxies

While surface brightness profiles, such as the generalized de Vaucouleurs profile, sample the distribution of luminous matter in a galaxy, they do not reveal the distribution of the galaxy's dark matter. A direct means to determine the distribution of all matter, luminous and dark, is to measure the galaxy's rotation curve, as was discussed for the case of the Milky Way in Chapter 24. Examples of rotation curves, averaged over groups of galaxies of a given M_B range and Hubble type, are shown in Fig. 25.8. You might also refer again to Fig. 24.27, which shows actual rotation curves for galaxies of various Hubble types.

When rotation curves are compared with either luminosity or Hubble type, a number of correlations are found. For instance, with increasing luminosity in the B band, L_B , the rotation curves tend to rise more rapidly with distance from the center and peak at higher maximum velocities (V_{max}). For galaxies of equal B -band luminosities, spirals of earlier type have larger values of V_{max} . Within a given Hubble type, galaxies that are more luminous have larger values of V_{max} . Also, for a given value of V_{max} , the rotation curves tend to rise slightly more rapidly with radius for galaxies of progressively earlier type. The fact that galaxies of different Hubble types, and therefore very different bulge-to-disk luminosity ratios, exhibit rotation curves that are very similar in form if not in amplitude suggests that the shapes of their gravitational potentials do not necessarily follow the distribution of luminous matter. This behavior is believed to be a signature of the existence of dark matter in these galaxies.

Although the maximum rotational velocity within the disk increases for earlier-type galaxies, a wide range in V_{max} exists for each type. For typical samples of spirals of type Sc and earlier, the mean maximum rotation velocities are $V_{\text{max}} = 299 \text{ km s}^{-1}$ for Sa, 222 km s^{-1} for Sb, and 175 km s^{-1} for Sc, while the ranges in values are 163 km s^{-1} to

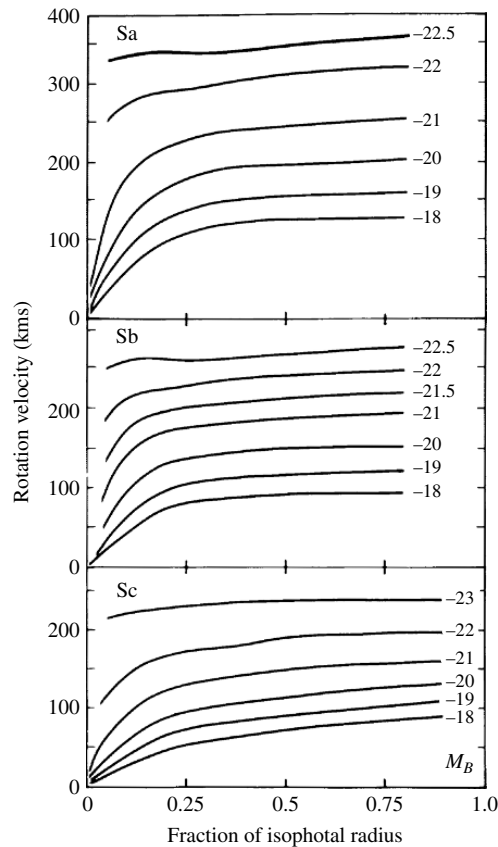


FIGURE 25.8 Variations in the average rotation curves of Sa, Sb, and Sc galaxies for various values of absolute magnitude in the B band. (Figure adapted from Rubin et al., *Ap. J.*, 289, 81, 1985.)

367 km s^{-1} , 144 km s^{-1} to 330 km s^{-1} , and 99 km s^{-1} to 304 km s^{-1} , respectively. Notice that the value of $V_{\text{max}} \simeq 250 \text{ km s}^{-1}$ for the Milky Way Galaxy (believed to be an SBbc; see Fig. 24.25) is only slightly greater than the mean value of V_{max} for Sb's.

The corresponding maximum rotation velocities for irregular galaxies is significantly lower than it is for earlier-type spirals, typically ranging from 50 to 70 km s^{-1} . This seems to suggest that a minimum rotation speed of roughly 50 to 100 km s^{-1} may be required for the development of a well-organized spiral pattern. The slower rotation velocities of Im's imply that their values of the rotational angular momentum per unit mass are only about 10% of the value found for our Galaxy in the solar neighborhood.

The Tully–Fisher Relation

As the preceding discussion suggests, a relationship exists between the luminosity of a spiral galaxy and its maximum rotation velocity. This correlation, now known as the **Tully–Fisher relation**, was first determined in 1977 by R. Brent Tully and J. Richard Fisher when

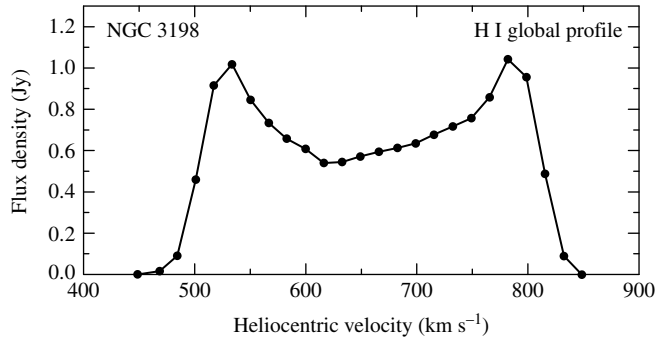


FIGURE 25.9 The global H I profile for NGC 3198. (Data from Begeman, *Astron. Astrophys.*, 223, 81, 1991.)

they measured the Doppler-broadened 21-cm radio emission line of neutral hydrogen in a sample of spiral galaxies. When the 21-cm line is sampled across the entire galaxy at one time, this line typically displays a double peak, as shown in Fig. 25.9. The double peak arises because of the flat rotation curve of the galaxy, which generally has the highest rotational velocity in the flat part of the curve. Since so much of the H I participates in the rotation at this maximum velocity, the flux density is greatest at this value. The double peak occurs because a portion of the disk is rotating toward the observer, causing the line to be blueshifted, and a portion of the disk is rotating away from the observer, resulting in a redshifted line. The average radial velocity of the galaxy relative to the observer is the midpoint value between the two peaks.

The shift $\Delta\lambda$ of a peak from its rest wavelength is given by Eq. (4.39),

$$\frac{\Delta\lambda}{\lambda_{\text{rest}}} \simeq \frac{v_r}{c} = \frac{V \sin i}{c}.$$

Here, v_r is the radial velocity, and i is the angle of inclination between the observer's line of sight and the direction perpendicular to the galactic plane (so $i = 90^\circ$ when viewing the galaxy edge-on).

Figure 25.10 shows the Tully–Fisher relation as M_B vs. V_{max} for a sample of Sa, Sb, and Sc galaxies. Notice the shift to lower values of V_{max} for galaxies of later Hubble type but with similar M_B 's. When the data are fitted with linear relations that depend on Hubble type, we find that

$$M_B = -9.95 \log_{10} V_{\text{max}} + 3.15 \quad (\text{Sa}), \quad (25.5)$$

$$M_B = -10.2 \log_{10} V_{\text{max}} + 2.71 \quad (\text{Sb}), \quad (25.6)$$

$$M_B = -11.0 \log_{10} V_{\text{max}} + 3.31 \quad (\text{Sc}). \quad (25.7)$$

The Tully–Fisher relation can be further refined and tightened if observations are made at infrared wavelengths. This offers two advantages. Observing at dust-penetrating infrared

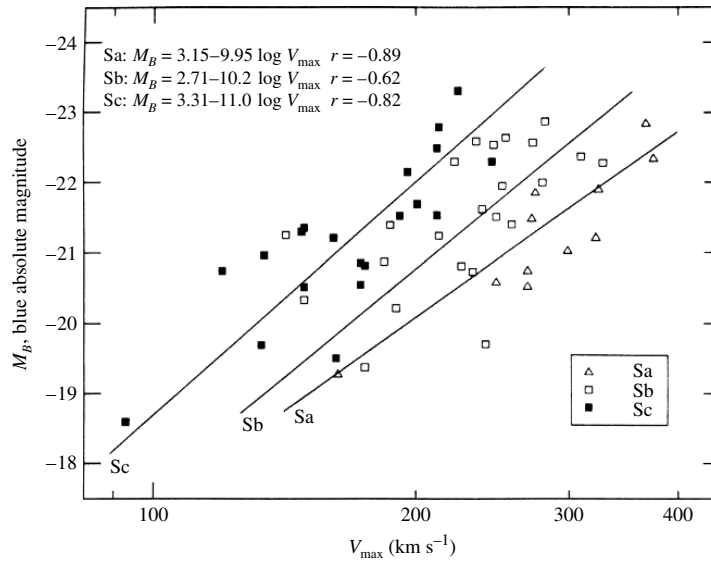


FIGURE 25.10 The Tully–Fisher relation for early spiral galaxies. (Figure adapted from Rubin et al., *Ap. J.*, 289, 81, 1985.)

wavelengths reduces extinction by a factor of 10. In addition, the infrared light comes primarily from late-type giant stars that are good tracers of the galaxy’s overall luminous mass distribution; the B band tends to emphasize young, hot stars in regions of recent star formation. One expression of the Tully–Fisher relation in the infrared H wavelength band ($1.66 \mu\text{m}$), devised by Pierce and Tully (1992), is

$$M_H^i = -9.50(\log_{10} W_R^i - 2.50) - 21.67 \pm 0.08. \quad (25.8)$$

W_R^i is a measure of the rotation of the galaxy defined as

$$W_R^i \equiv (W_{20} - W_{\text{rand}}) / \sin i, \quad (25.9)$$

where W_{20} is the velocity difference between the blueshifted and redshifted emission in the H band when the intensity of the emission is 20% of its blue and red peak values. W_{rand} is a measure of the random velocities superimposed on observed velocities due to noncircular orbital motions in the galaxy.⁹ Finally, i is the inclination angle of the plane of the galaxy. An example of the H -band Tully–Fisher relation (Eq. 25.8) for galaxies in three clusters is shown in Fig. 25.11.

Although the exact form of the Tully–Fisher relation depends on the distribution of mass within galaxies, as well as on variations in their mass-to-light ratios, we can still gain some insight into its origin. As is evident in Figs. 24.27 and 25.8, spiral galaxies have nearly

⁹For a Gaussian random velocity distribution, it can be shown that W_{rand} is related to the velocity dispersion σ by $W_{\text{rand}} = 3.6\sigma$.

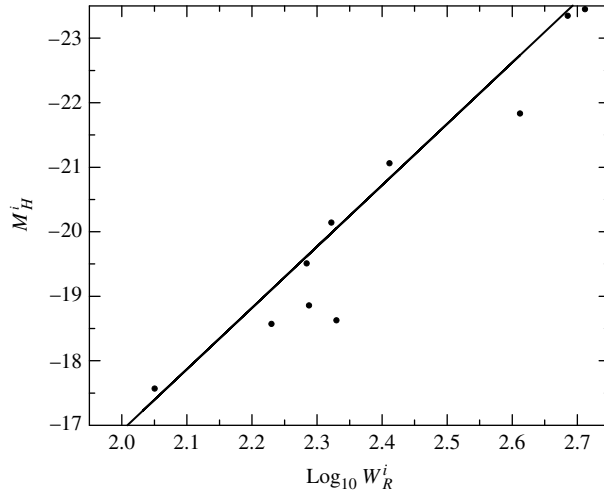


FIGURE 25.11 The Tully–Fisher relation in the infrared H -band ($1.66 \mu\text{m}$) as given by Eq. (25.8). The data are for galaxies in the Local, Sculptor, and M81 groups of galaxies. (Data from Pierce and Tully, *Ap. J.*, 387, 47, 1992.)

flat rotation curves beyond a few kiloparsecs. In Section 24.3 we saw that this implies that the mass contained within a radius r is given by Eq. (24.49). Evaluating this for the entire galaxy ($r \rightarrow R$ and $M_r \rightarrow M$) results in

$$M = \frac{V_{\max}^2 R}{G}, \quad (25.10)$$

where the maximum rotation speed, $V_{\max}$, is equated with the flat portion of the rotation curve. Now, *if* the mass-to-light ratio has the same value for all spirals ($M/L \equiv 1/C_{\text{ML}}$, where C_{ML} is a constant), then

$$L = C_{\text{ML}} \frac{V_{\max}^2 R}{G}.$$

Finally, if we make the crude assumption that all spirals have the same surface brightness at their centers, then $L/R^2 \equiv C_{\text{SB}}$, where C_{SB} is another constant. Eliminating R from the expression for the luminosity, we obtain

$$L = \frac{C_{\text{ML}}^2}{C_{\text{SB}}} \frac{V_{\max}^4}{G^2} = C V_{\max}^4,$$

where C incorporates the other constants. The absolute magnitude comes from Eq. (3.8),

$$\begin{aligned} M &= M_{\text{Sun}} - 2.5 \log_{10} \left(\frac{L}{L_{\odot}} \right) \\ &= M_{\text{Sun}} - 2.5 \log_{10} V_{\max}^4 - 2.5 \log_{10} C + 2.5 \log_{10} L_{\odot} \\ &= -10 \log_{10} V_{\max} + \text{constant}. \end{aligned}$$

Although the additive (zero-point) constant remains unevaluated, this simple argument nearly reproduces the leading coefficients of Eqs. (25.5–25.8), the slopes of the B -band Tully–Fisher relations for Sa’s, Sb’s, and Sc’s, and the slope of the collective relationship in the H band.

Radius–Luminosity Relation

Another important pattern also emerges in the data of early-type spiral galaxies (e.g., Sa–Sc): Radius increases with increasing luminosity, independent of Hubble type. At the disk radius (R_{25}) corresponding to a surface-brightness level of 25 B -mag arcsec $^{-2}$, the data are well represented by the linear relationship

$$\log_{10} R_{25} = -0.249M_B - 4.00, \quad (25.11)$$

where R_{25} is measured in units of kpc.

Masses and Mass-to-Light Ratios

Through the use of the Tully–Fisher relation (Eqs. 25.5–25.7), combined with Eqs. (25.11) and (24.49), it is now possible to estimate both the masses of early-type spiral galaxies and their mass-to-light ratios interior to R_{25} . While masses range from greater than $10^9 M_\odot$ to at most about $10^{12} M_\odot$, there is only a very weak relationship between a galaxy’s mass and its Hubble type (on average, Sc’s are *slightly* less massive than Sa’s), as might be expected, given the lack of any dependence on Hubble type in the correlation between R_{25} and M_B . However, a correlation does exist for the mass-to-light ratios, with the average value of M/L_B decreasing with progressively later Hubble type; $\langle M/L_B \rangle = 6.2 \pm 0.6$ for Sa’s, 4.5 ± 0.4 for Sb’s, and 2.6 ± 0.2 for Sc’s.

Colors and the Abundance of Gas and Dust

The trend in M/L_B suggests that Sc’s tend to have a greater fraction of massive main-sequence stars relative to earlier spirals (recall from the mass–luminosity relation shown in Fig. 7.7 that upper main-sequence stars have low mass-to-light ratios). If this is the case, we should also expect Sc’s to be bluer than Sa’s and Sb’s, which is just what is observed. The mean values of the color index, $B - V$, decrease with later Hubble types: 0.75 for Sa’s, 0.64 for Sb’s, and 0.52 for Sc’s. For successively later-type galaxies, progressively greater portions of the overall light from spirals is emitted in bluer wavelength regions, implying an increasingly greater fraction of younger, more massive, main-sequence stars.¹⁰

Irregulars tend to be the bluest of all galaxies represented by the Hubble sequence, with characteristic values of $B - V \sim 0.4$. Furthermore, Ir’s often get bluer toward their centers, rather than redder as is the case for early-type spirals. This suggests that irregulars are still actively manufacturing stars in their central regions. For instance, the LMC and the SMC still appear to be making blue globular clusters in their disks.

¹⁰It is an unfortunate consequence of the historical development of the terminology that galaxies of later Hubble type are actually dominated by stars lying on the early (upper) part of the main sequence.

Since blue main-sequence stars are short-lived, they must have formed relatively recently. Presumably an abundant supply of gas and dust exists in Sc's from which these stars can be produced. Based on the analysis of 21-cm radiation, H α emission, and CO emission (which is a good tracer of H₂), we find that the mass fraction of gas relative to the total mass interior to R_{25} increases steadily from Sa's to Sc's and later. For instance, $\langle M_{\text{gas}}/M_{\text{total}} \rangle = 0.04$ for Sa's, 0.08 for Sb's, 0.16 for Sc's, and 0.25 for Scd's. This agrees with the observation that the brightness of H II emission also increases with later Hubble types.

The relative amounts of atomic and molecular hydrogen also change with Hubble type. For Sab's, $\langle M_{\text{H}_2}/M_{\text{H I}} \rangle = 2.2 \pm 0.6$, decreasing to 1.8 ± 0.3 for Sb's, 0.73 ± 0.13 for Sc's, and 0.29 ± 0.07 for Scd's. This observation has been interpreted as implying that Sa's are somewhat more centrally condensed, containing correspondingly deeper gravitational wells in which gas can collect and combine to form molecules. This means that the interstellar medium of Sa's is dominated by molecular gas while the interstellar medium of Scd's is primarily composed of atomic hydrogen. Overall, the amount of molecular hydrogen in spirals can range from $5 \times 10^{10} M_{\odot}$ for the most massive galaxies to as little as $10^6 M_{\odot}$ for dwarf spirals. The mass of dust is characteristically 150 to 600 times lower than the mass of gas in the ISM.

It is dust that is primarily responsible for the far-infrared (FIR) luminosities of galaxies, although synchrotron radiation and the emissions of stars can also contribute to the total. From observations made by IRAS, astronomers have discovered that $L_{\text{FIR}}/L_B = 0.07$ for M31 (an SbI–II), 0.2 for M33 (an Sc(s)II–II), 0.4 for M101 (an Sc(s)I), 0.18 for the LMC (an SBmIII), and 0.09 for the SMC (an ImIV–V). In general, Sc's have a larger fraction of infrared emission than Sa's or Sb's, consistent with the other observations of correspondingly greater fractions of their masses being in the form of gas and dust. Interestingly, SB's tend to have greater infrared luminosities than do normal S's.

Metallicity Gradients and Color Gradients of Spirals

Not only is there a dependence of color on Hubble type, but individual spiral galaxies also exhibit *color gradients*, with their bulges generally being redder than their disks. This arises for two reasons: metallicity gradients and star formation activity. The average number of electrons per atom is larger for metal-rich stars than for metal-poor stars. Since ionization and the orbital transitions of electrons contribute to the opacity in stellar photospheres, the opacity in metal-rich stars is greater. Because it is more difficult for light generated in the interior to escape from a star with a higher-opacity photosphere, the star will tend to “puff up”; its radius will increase, with a corresponding decrease in effective temperature. Hence, a higher-opacity star will be redder than a lower-opacity star, all else being equal (this effect was first discussed in Example 9.5.4 of Section 9.5). The redness of bulges argues for those regions being more metal-rich than are the portions of disks farther from the center, as is the case in our own Galaxy. In fact, within the Milky Way, metallicity gradients have been measured at Galactocentric radii of between 4 and 14 kpc with values of $d[\text{He}/\text{H}]/dr = -0.01 \pm 0.008 \text{ dex kpc}^{-1}$, $d[\text{O}/\text{H}]/dr = -0.07 \pm 0.015 \text{ dex kpc}^{-1}$, and $d[\text{Fe}/\text{H}]/dr = -0.01 \text{ to } -0.05 \text{ dex kpc}^{-1}$.¹¹

¹¹“dex” refers to the logarithmic nature of the metallicity term in the gradient; recall Eq. (24.11).

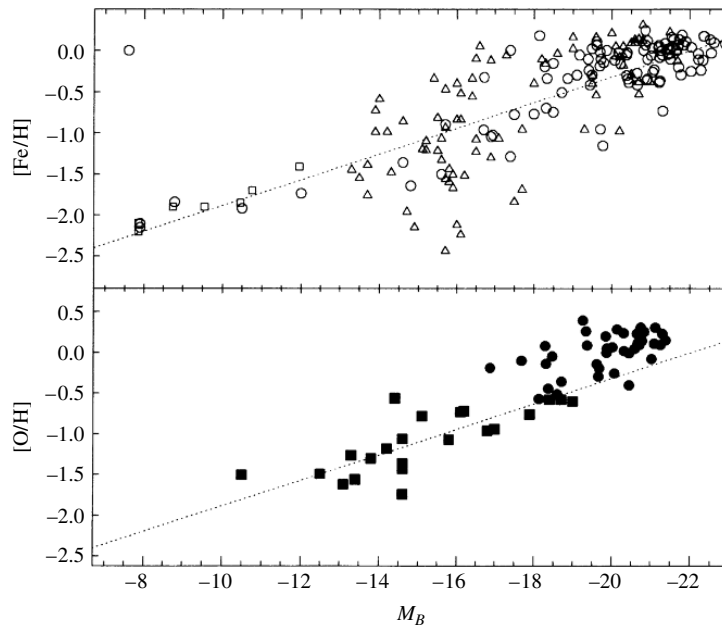


FIGURE 25.12 Metallicity is observed to increase linearly with absolute magnitude. The symbols in the upper panel correspond to dE's (dwarf ellipticals, open squares) and ellipticals (open circles and open triangles). In the lower panel, spirals are represented by filled circles and irregulars are indicated by filled squares. (Figure adapted from Zaritsky, Kennicutt, and Huchra, *Ap. J.*, 420, 87, 1994.)

Star formation, the second major cause of color gradients, implies that the disks of spiral galaxies are more actively involved in star-making than are their bulges. This is consistent with the distribution of gas and dust in the galaxies. The spheroidal components of spirals usually contain much less gas and have correspondingly lower star formation rates than their disks. Since disks are able to produce young, hot, blue stars at a greater rate, the spheroids appear relatively redder and a color gradient is established.

Color gradients have been observed within the spheroids of spiral galaxies as well, such as NGC 7814. The galaxies' spheroidal components become bluer with increasing radius. This is also the case for the Milky Way, with the more metal-rich, redder globular clusters orbiting closer to the Galactic center.

Observations indicate that metallicity correlates with the absolute magnitudes of galaxies; $[\text{Fe}/\text{H}]$ and $[\text{O}/\text{H}]$ both increase with M_B (see Fig. 25.12). Apparently chemical enrichment was somehow more efficient in luminous, massive galaxies. Composition enrichment histories and gradients have significant implications for galaxy formation theories, as will be discussed in Section 26.2.

X-Ray Luminosity

The X-ray luminosities of galaxies also provide some information concerning their evolution. In spirals, luminosities in the wavelength region sampled by the Einstein satellite (with photon energies of 0.2 to 3.5 keV) typically range from $L_X = 10^{31}$ W to 10^{34} W. A sur-

prisingly tight correlation exists between X-ray and B -band luminosities ($L_X/L_B \simeq 10^{-7}$), which has been interpreted as implying that the X-rays are due to a class of objects that constitutes an approximately constant fraction of the population of all objects in spirals. The suspected sources are X-ray binaries. It is probable that supernova remnants also contribute to the X-ray emission.

Supermassive Black Holes

Observations of stellar and gas motions near the centers of some spirals strongly suggest the presence of supermassive black holes. For instance, near the center of M31, M/L exceeds $35 M_\odot/L_\odot$, indicating a large amount of nonluminous matter confined to a small region. Rotational-velocity measurements can be used to estimate the *dynamical mass* of the central black hole of M31 in the same way it was done for Sgr A* in the Galactic center (see Section 24.4). A precise determination based on kinematic studies of the triple nucleus of M31 (Fig. 25.13) gives a mass of $1.4^{+0.9}_{-0.3} \times 10^8 M_\odot$ for the central supermassive black hole.

Another (although less precise) method of determining the mass of a central supermassive black hole uses the velocity dispersion to obtain a mass estimate via the virial theorem. According to Eq. (2.44), the time-averaged kinetic and potential energies of stars in the galaxy's central region are related by

$$\frac{1}{2} \left\langle \frac{d^2 I}{dt^2} \right\rangle - 2 \langle K \rangle = \langle U \rangle ,$$

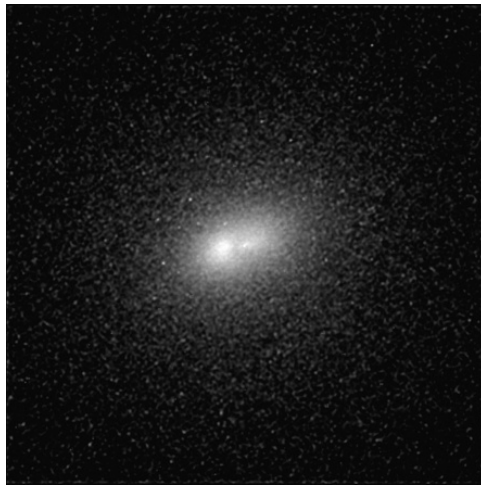


FIGURE 25.13 An HST image of the triple nucleus of the Andromeda galaxy (M31). Two groupings of red stars are evident in this image. The brighter grouping of red stars on the left is P1, and the dimmer grouping is P2. P1 and P2 are believed to be portions of an elongated disk of stars orbiting the galaxy's supermassive black hole. P1 is farthest from the black hole (the apocenter), and P2 is nearest the black hole (the pericenter). A blue cluster of stars, known as P3, is superimposed on P2. P3 is believed to be orbiting the galaxy's supermassive black hole. [Courtesy of NASA, ESA, and T. Lauer (NOAO/AURA/NSF).]

where I is the region's moment of inertia. If the galaxy is in equilibrium, then $\langle d^2 I / dt^2 \rangle = 0$, resulting in the usual statement of the virial theorem,

$$-2 \langle K \rangle = \langle U \rangle.$$

Furthermore, for a large number of stars, the central bulge will look the same (in a statistical sense) at any time, and the time-averaging can be dropped. So for N stars,

$$-2 \sum_{i=1}^N \frac{1}{2} m_i v_i^2 = U,$$

where U is given by Eq. (2.43).

For simplicity, we restrict our attention to a spherical cluster of radius R with N stars, each of mass m , so the total mass of the bulge is $M = Nm$. Dividing the above expression by N produces

$$-\frac{m}{N} \sum_{i=1}^N v_i^2 = \frac{U}{N}. \quad (25.12)$$

Of course, astronomers actually measure the radial component of the velocity vector (a galaxy is too far away to allow for detection of proper motions). Moreover, an astronomer is just as likely to see a star moving in the radial direction as in either of the other two perpendicular directions. With the brackets denoting an average value,¹²

$$\langle v^2 \rangle = \langle v_r^2 \rangle + \langle v_\theta^2 \rangle + \langle v_\phi^2 \rangle = 3 \langle v_r^2 \rangle,$$

so

$$\frac{1}{N} \sum_{i=1}^N v_i^2 = \langle v^2 \rangle = 3 \langle v_r^2 \rangle = 3\sigma_r^2,$$

where σ_r is the dispersion in the radial velocity (see Section 24.3). Inserting this result into Eq. (25.12), and using Eq. (10.22) for the (approximate) potential energy of a spherical distribution of total mass M and radius R , leads to

$$-3m\sigma_r^2 \approx -\frac{3}{5} \frac{GM^2}{NR}.$$

Using $M = Nm$ and solving for the mass give

$$\boxed{M_{\text{virial}} \approx \frac{5R\sigma_r^2}{G}}, \quad (25.13)$$

where the mass obtained in this way is called the **virial mass**.

¹²This procedure is identical to the one that was used in Section 10.2 to derive the pressure integral.

It is important to note that in order to determine an accurate value for the mass, an appropriate choice of R must be made. As the observations move farther from the black hole, contributions to the total mass increase from surrounding stars and gas. R must be chosen to be within the black hole's "sphere of influence."

Example 25.2.1. Equation (25.13) can be used to estimate a virial mass for the central black hole of M32, a companion galaxy of M31. From Fig. 25.14, the central radial-velocity dispersion is approximately 162 km s^{-1} . This means that within $0.1''$ (approximately 0.4 pc) of the center, there is a total mass of roughly

$$M_{\text{virial}} \sim 1 \times 10^7 M_{\odot}.$$

A more accurate estimate, based on the rotation curve, gives a value of between $1.5 \times 10^6 M_{\odot}$ and $5 \times 10^6 M_{\odot}$.

It is important to note that central supermassive black holes are not restricted to late-type galaxies. For example, based on observations made using the Hubble Space Telescope, it appears that the giant elliptical galaxy M87 (NGC 4476) also contains a $3.2 \pm 0.9 \times 10^9 M_{\odot}$ black hole. HST was able to resolve a disk of material within M87 that has rotational speeds reaching 550 km s^{-1} ; the disk itself is orbiting a central region no larger than our Solar System. Given that the distance to M87 is nearly 20 Mpc , this is a remarkable observation. (M87 is also known to have a relativistic jet that is believed to be powered by the supermassive black hole at its center; see Fig. 28.10.)

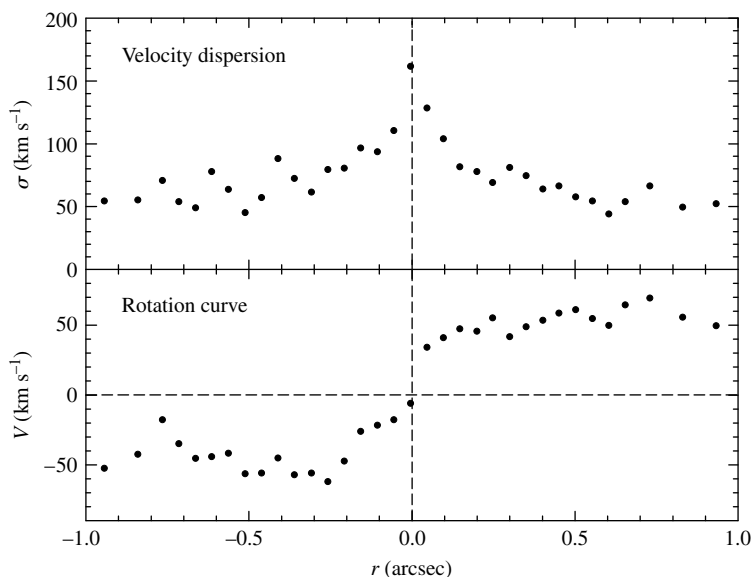


FIGURE 25.14 The stellar velocity dispersion and rotational velocities of stars near the center of M32. Given the distance to M32 of 770 kpc , $1''$ corresponds to a linear distance from the center of 3.7 pc . (Data from Joseph, et al., *Ap. J.*, 550, 668, 2001.)

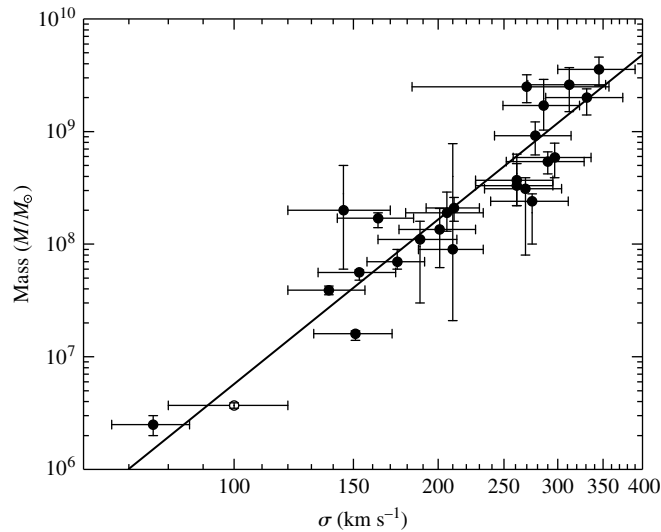


FIGURE 25.15 The relationship between the mass of a galaxy's supermassive black hole and the velocity dispersion, σ , of the galaxy's spheroid. The data set includes elliptical, lenticular, and spiral galaxies. (Data from Ferrarese and Ford, *Space Science Reviews*, 116, 523, 2005. The open symbol represents the Milky Way with mass data from Ghez, et. al, *Ap. J.*, 620, 744, 2005.)

As the number of known central supermassive black holes has increased, it has become possible to look for relationships between the black holes and their host galaxies. A very intriguing and useful correlation has been discovered between the mass of the supermassive black hole in a galaxy's center and the velocity dispersion of the stars within the galaxy. The relationship is illustrated in Fig. 25.15 and is given by the best-fit power law,

$$M_{\text{bh}} = \alpha(\sigma/\sigma_0)^\beta, \quad (25.14)$$

where σ is the velocity dispersion in units of km s^{-1} , $\alpha = (1.66 \pm 0.24) \times 10^8 M_\odot$, $\beta = 4.86 \pm 0.43$, and $\sigma_0 = 200 \text{ km s}^{-1}$. The velocity dispersion is measured for the stellar population near the black hole. Correlations between the mass of supermassive black holes and other bulk galaxy parameters, such as the luminosity of the bulge, have also been uncovered. Apparently a fundamental link exists between the formation of the central supermassive black hole in a galaxy and the overall formation of the galaxy itself. Just what that link is remains to be determined. (Galaxy formation will be discussed in more detail in Chapter 26.)

Evidence continues to mount that supermassive black holes may exist in the centers of most galaxies. The dramatic activity associated with the central supermassive black holes of M87 and other galaxies is the subject of Chapter 28.

Specific Frequency of Globular Clusters

Finally, it is worth discussing the abundance of globular clusters in late-type galaxies, which, when compared with the numbers of globular clusters in ellipticals, has important implications for theories of galactic formation and evolution. It appears that galaxies that

are more spheroidally dominant (i.e., earlier Hubble types) were more efficient at forming globular clusters during their early histories. Although virtually all galaxies appear to contain some globular clusters, the number of clusters within a galaxy seems to increase with total galaxy luminosity and with progressively earlier Hubble type.

To compare cluster systems among Hubble types more directly, the globular-cluster counts are usually normalized to a standard absolute magnitude for the parent galaxy of $M_V = -15$ mag (in the visual wavelength band). If N_t is the total number of globular clusters in a galaxy, then the **specific frequency** of globular clusters is defined as

$$\begin{aligned} S_N &= N_t \frac{L_{15}}{L_V} \\ &= N_t 10^{0.4(M_V + 15)}, \end{aligned} \quad (25.15)$$

where L_V is the galaxy's luminosity and L_{15} is the reference luminosity corresponding to an absolute visual magnitude of $M_V = -15$ mag [Eq. (3.7) was used to arrive at the second expression]. As can be seen in Fig. 25.16, for Sc galaxies and later, the average value of S_N is in the range of 0.5 ± 0.2 , while for Sa's and Sb's, $\langle S_N \rangle$ increases to 1.2 ± 0.2 . $\langle S_N \rangle$ is even larger for elliptical galaxies, meaning that they have more clusters per unit luminosity than do spirals. In fact, by far the largest number of globular clusters per unit luminosity belongs to the gigantic elliptical galaxies (cD's) that are frequently found near the centers of large clusters of galaxies (see Sections 25.4 and 27.3). It seems that the number of globular clusters offers important clues to galaxy formation theories.

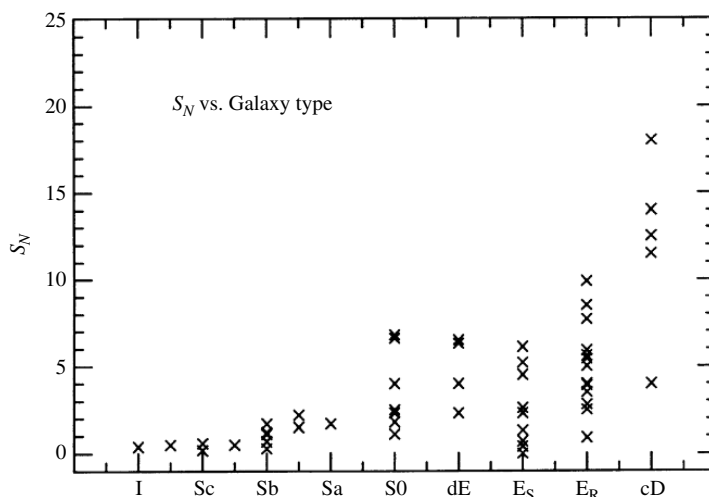


FIGURE 25.16 The association of the specific frequency of globular clusters with Hubble type. cD galaxies are the largest elliptical galaxies in the universe; dE's are dwarf ellipticals. (Figure adapted from Harris, *Annu. Rev. Astron. Astrophys.*, 29, 543, 1991. Reproduced by permission from the *Annual Review of Astronomy and Astrophysics*, Volume 29, ©1991 by Annual Reviews Inc.)

25.3 ■ SPIRAL STRUCTURE

Galaxies exhibit a rich variety of spiral structure, which may vary in number of arms and how tightly wound they are, degree of smoothness in the distribution of stars and gas, surface brightness, and the existence or lack of bars. The most majestic spiral galaxies, known as **grand-design spirals**, usually have two very symmetric and well-defined arms. One of the best-known examples of a grand-design spiral is M51 (NGC 5194), shown in Fig. 25.17. Also called the Whirlpool galaxy, M51 has a companion galaxy (NGC 5195, visible in the image) near the end of one of the spiral arms.

However, not all spirals are grand designs with two distinct arms. For instance, M101 [Fig. 25.5(c)] has four arms, and NGC 2841 (Fig. 25.18) has a series of partial arm fragments. Galaxies like NGC 2841, which do not possess well-defined spiral arms that are traceable over a significant angular distance, are called **flocculent spirals**. Only about 10% of all spirals are considered grand-design galaxies, another 60% are multiple-arm galaxies, and the remaining 30% are flocculent galaxies.

The optical images of spiral galaxies are dominated by their arms. This is because very luminous O and B main-sequence stars and H II regions are found preferentially in the arms. Since massive OB stars are short-lived objects relative to the characteristic rotation period of a galaxy, spiral structure must correspond to regions of active star formation. (For instance, the ages of OB stars are on the order of 10 Myr, compared to 230 Myr for the orbital period of the local standard of rest in the Milky Way; see Table 13.1 and Example 24.3.1, respectively.)



FIGURE 25.17 The Whirlpool galaxy, M51 (NGC 5194), is an Sbc(s)I–II grand-design spiral located in the constellation Canes Venatici (the Hunting Dogs of Boötes, just below Ursa Major). Also visible is its companion NGC 5195, situated near the end of one of the spiral arms. (Image from Sandage and Bedke, *The Carnegie Atlas of Galaxies*, Carnegie Institution of Washington, Washington, D.C., 1994.)



FIGURE 25.18 The Sb galaxy NGC 2841 is an example of a flocculent spiral. (Image from Sandage and Bedke, *The Carnegie Atlas of Galaxies*, Carnegie Institution of Washington, Washington, D.C., 1994.)

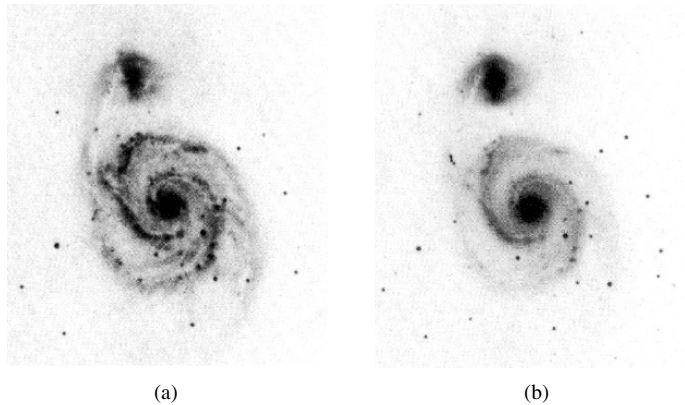


FIGURE 25.19 M51 as seen in (a) blue light and (b) red light. (Figures from Elmegreen, *Ap. J. Suppl.*, 47, 229, 1981.)

Careful inspection of the various images found in this chapter reveals that dust bands are also evident in the spiral arms; M51 (Fig. 25.17) is a particularly good example. Notice that the dust bands tend to reside on the inner (concave) edges of the arms. Observation of 21-cm H I emission indicates that gas clouds are also more prevalent near the inner edges of the arms.

As Fig. 25.19 illustrates, when spiral galaxies are observed in red light, the arms become much broader and less pronounced, although they still remain detectable. Since observations

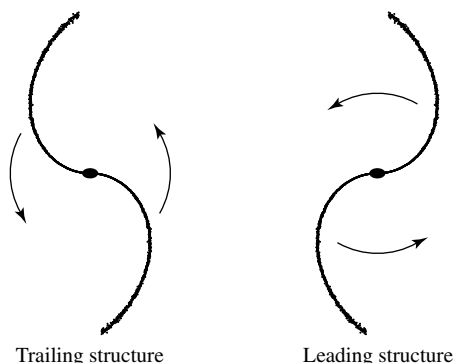


FIGURE 25.20 Trailing and leading spiral arm structures.

at red wavelengths emphasize the emission of long-lived, lower-mass main-sequence stars and red giants, this implies that the bulk of the disk is dominated by older stars. However, despite the dominance of lower-mass stars between the spiral arms, observations indicate that there is still an increase in the number density of older stars (and an accompanying increase in mass density) within the spiral arms.

Trailing and Leading Spiral Arms

Although the general appearance of spiral galaxies suggests that their arms are *trailing*, meaning that the tips of the arms point in the opposite direction from the direction of rotation (see Fig. 25.20), verifying this is not always a simple matter. Distinguishing between trailing and *leading* spiral arms requires a determination of the orientation of the plane of the galaxy relative to our line of sight so that radial-velocity measurements can be unambiguously interpreted in terms of the direction of galaxy rotation. In almost all cases where such a clear determination can be made, it does appear that spiral arms are trailing. However, in one case, NGC 4622, two arms are going one way and another arm is winding in the opposite direction; at least one of these arms must be leading. It has also been suggested that M31 (Andromeda) has one tightly wound leading arm. In each case it is likely that the cause of the leading spiral is a tidal encounter with a retrograde-moving object (M32 in the case of Andromeda).

The Winding Problem

Given that spiral galaxies are commonplace within the universe, it is natural to ask what causes spiral structure, and whether spiral arms are long-lived (with lifetimes comparable to the age of a galaxy) or transient.

One problem immediately arises when the nature of spiral structure is considered; *material arms* composed of a fixed set of identifiable stars and gas clouds would necessarily “wind up” on a timescale that is short compared to the age of the galaxy. This so-called **winding problem** can be understood by considering a set of stars that are originally along a single line, but at varying distances from the center of the galaxy, as shown in Fig. 25.21(a). Since the disk of a spiral galaxy rotates differentially (except very near the center), the outer

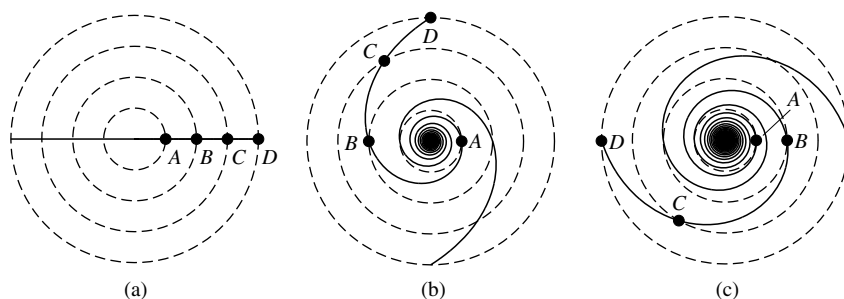


FIGURE 25.21 The winding problem for material arms. The arms become progressively more tightly wound as time goes on. A flat rotation curve is assumed with $R_B = 2R_A$, $R_C = 3R_A$, and $R_D = 4R_A$. (a) The stars start out in a line at time $t = 0$. (b) After star A has completed one orbit. (c) After star A has completed two orbits.

stars will require more time to complete one orbit than will stars with smaller orbital radii.¹³ This effect will of course lead to a natural generation of trailing spiral arms. However, after only a few orbits, the spiral arms will become too tightly wound to be observed; this situation is depicted in Figs. 25.21(b) and 25.21(c). Another mechanism is needed to explain persistent spiral structure.

The Lin–Shu Density Wave Theory

In the mid-1960s, the American astronomers C. C. Lin and Frank Shu proposed that spiral structure arises because of the presence of long-lived **quasistatic density waves**. Density waves consist of regions in the galactic disk where the mass density is greater than average, perhaps by 10% to 20%. Stars, dust, and gas clouds move through the density waves during their orbits around the galactic center, much like cars slowly working their way through a traffic jam on a highway. Lin and Shu suggested that when the galaxy is viewed in a noninertial reference frame that is rotating with a specific angular speed Ω_{gp} , known as the **global pattern speed**, the spiral wave pattern appears to be stationary, as shown in Fig. 25.22. This does not imply that the motions of the stars are also stationary in that frame, however. Stars near the center of the galaxy can have orbital periods that are shorter than the density wave pattern (or $\Omega > \Omega_{gp}$) and so they will overtake a spiral arm, move through it, and continue on until they encounter the next arm. Stars sufficiently far from the center of the galaxy will be moving more slowly than the density wave pattern and will be overtaken by it ($\Omega < \Omega_{gp}$). At a specific distance from the center, called the **corotation radius** (R_c), the stars and density waves will move together. In this noninertial reference frame in which the density wave pattern is static, the stars with $R < R_c$ will appear to pass through the arms moving in one direction, while stars with $R > R_c$ will appear to be moving in the opposite sense.

The Lin–Shu hypothesis helps to explain many of the observations concerning spiral structure that have been discussed—for instance, the ordering of H I clouds and dust bands

¹³For flat rotation curves, the orbital velocities of stars are nearly independent of distance from the center, but the *angular velocity*, $\Omega = v/R$, still decreases with increasing distance.

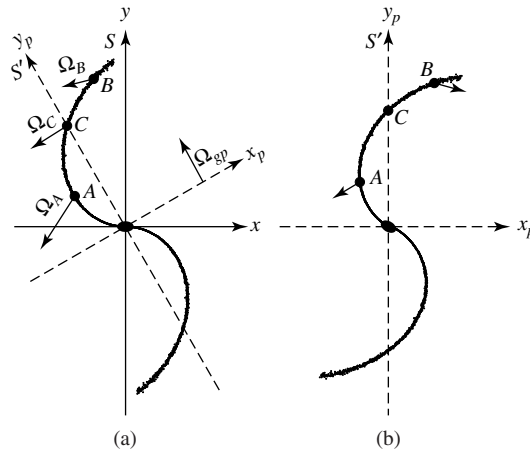


FIGURE 25.22 (a) A galaxy with trailing spiral arms as seen in an inertial reference frame S in which a quasistatic density wave is moving with a global angular pattern speed Ω_{gp} . Star A has an angular speed $\Omega_A > \Omega_{gp}$, Star B has an angular speed $\Omega_B < \Omega_{gp}$, and Star C is corotating with the density wave, meaning that $\Omega_C = \Omega_{gp}$. (b) The motions of the stars as seen in a noninertial reference frame S' , corotating with the density wave.

on the inner (trailing) edges of spiral arms, the existence of young, massive stars and H II regions throughout the arms, and an abundance of old, red stars in the remainder of the disk. Apparently, as dust and gas clouds within the corotation radius overtake a density wave, they are compressed by the effects of the increase in local mass density. This causes some of the clouds to satisfy the Jeans criterion (Eqs. 12.14 and 12.16) and begin to collapse, resulting in the formation of new stars. Since this process takes some time (approximately 10^5 yr for a $15 M_{\odot}$ star; see Table 12.1), the appearance of new stars will occur within the arm slightly “downstream” from dust and gas clouds at the edge of the wave. The birth of the brightest and bluest new stars, the massive O and B stars, will result in the creation of H II regions as the UV ionizing radiation moves through the interstellar medium. Because massive stars have relatively short lifetimes, they will die before they can move entirely out of the density wave in which they were born. Less massive, redder stars will be able to live much longer (some longer than the current age of the galaxy) and so will continue through the density wave and become distributed throughout the disk. Local maxima in the number density of red dwarfs within spiral arms are due to the presence of the density wave during a subsequent passage, which causes the stars to collect at the bottom of the wave’s gravitational potential well. Of course, the same scenario could also occur on the outer (leading) edges of spiral arms outside the corotation radius. However, it is likely that less dust and gas will be found in these outer regions of the galaxy.

In principle, the density wave theory also suggests a solution to the winding problem. The problem arose because we considered material arms (arms composed of a fixed set of stars). If, instead, the stars are allowed to pass through a quasistatic density wave, then the problem has been changed to one of establishing and maintaining the wave of enhanced density. This has been the focus of considerable research since the Lin–Shu hypothesis was proposed.

Small-Amplitude Orbital Perturbations

We now turn our attention to developing the basic ideas of how the orbital motions of individual stars about the galactic center can result in spiral-shaped regions of enhanced density according to the Lin–Shu hypothesis. We begin by considering the general motion of a star (or gas cloud) in an axially symmetric gravitational field that is also symmetric about the galactic midplane. This implies that we are assuming here that density waves make an insignificant contribution to the gravitational field, an assumption that may not be valid in all spiral galaxies. However, the assumption does simplify the analysis considerably.

From Fig. 25.23, the position of a star at some general point above the galactic midplane can be written in the form

$$\mathbf{r} = R\hat{\mathbf{e}}_R + z\hat{\mathbf{e}}_z, \quad (25.16)$$

where $\hat{\mathbf{e}}_R$, $\hat{\mathbf{e}}_\phi$, and $\hat{\mathbf{e}}_z$ are the unit vectors in cylindrical coordinates.¹⁴ To convert between rectangular and cylindrical coordinates, we see that

$$x = R \cos \phi, \quad y = R \sin \phi, \quad z = z,$$

and

$$\hat{\mathbf{e}}_R = \hat{\mathbf{i}} \cos \phi + \hat{\mathbf{j}} \sin \phi, \quad \hat{\mathbf{e}}_\phi = -\hat{\mathbf{i}} \sin \phi + \hat{\mathbf{j}} \cos \phi, \quad \hat{\mathbf{e}}_z = \hat{\mathbf{k}}.$$

For a star of mass M , Newton's second law of motion in cylindrical coordinates is

$$M \frac{d^2 \mathbf{r}}{dt^2} = \mathbf{F}_g(R, z),$$

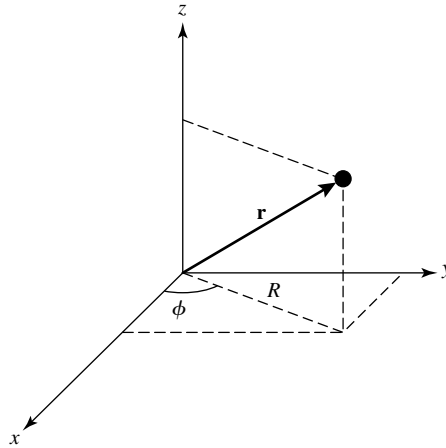


FIGURE 25.23 The position vector of a star in cylindrical coordinates. The origin of the coordinate system is at the center of the galaxy, and the x – y plane is in the galaxy's midplane.

¹⁴Note that we have chosen to use a traditional right-handed coordinate system here, rather than the left-handed one that we used to analyze the kinematics of the Milky Way Galaxy in Section 24.3. To avoid confusion, we will use ϕ rather than θ to designate the azimuthal angle.

where $\mathbf{F}_g$ is the gravitational force on the star, and the ϕ -dependence is neglected due to the assumption of axial symmetry. Replacing the force with the negative of the gradient of the gravitational potential energy (recall the discussion in Section 2.2), yields

$$M \frac{d^2 \mathbf{r}}{dt^2} = -\nabla U(R, z) = -\frac{\partial U}{\partial R} \hat{\mathbf{e}}_R - \frac{1}{R} \frac{\partial U}{\partial \phi} \hat{\mathbf{e}}_\phi - \frac{\partial U}{\partial z} \hat{\mathbf{e}}_z.$$

Dividing both sides of the last expression by the star's mass, writing the result in terms of the gravitational potential¹⁵ $\Phi \equiv U/M$, and noting that U is independent of ϕ give

$$\frac{d^2 \mathbf{r}}{dt^2} = -\frac{\partial \Phi}{\partial R} \hat{\mathbf{e}}_R - \frac{\partial \Phi}{\partial z} \hat{\mathbf{e}}_z. \quad (25.17)$$

It is left as an exercise (Problem 25.13) to show that the star's acceleration vector (the left-hand side of Eq. 25.17) can be written in the form

$$\frac{d^2 \mathbf{r}}{dt^2} = (\ddot{R} - R\dot{\phi}^2) \hat{\mathbf{e}}_R + \frac{1}{R} \frac{\partial (R^2 \dot{\phi})}{\partial t} \hat{\mathbf{e}}_\phi + \ddot{z} \hat{\mathbf{e}}_z, \quad (25.18)$$

where the dots represent time derivatives. Comparing Eqs. (25.17) and (25.18), we see that the motion in each coordinate is given by

$$\ddot{R} - R\dot{\phi}^2 = -\frac{\partial \Phi}{\partial R}, \quad (25.19)$$

$$\frac{1}{R} \frac{\partial (R^2 \dot{\phi})}{\partial t} = 0, \quad (25.20)$$

$$\ddot{z} = -\frac{\partial \Phi}{\partial z}. \quad (25.21)$$

Because we have assumed axial symmetry, there is no component of the force vector in the $\hat{\mathbf{e}}_\phi$ direction and consequently no component of torque along the z -axis (recall that $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$). Therefore, the z component of the star's orbital angular momentum (L_z) must be constant since $\tau_z = dL_z/dt = 0$ (i.e., angular momentum is conserved throughout the star's motion). This is exactly what Eq. (25.20) implies, since $R^2 \dot{\phi} = Rv_\phi = L_z/M = \text{constant}$. Defining

$$J_z \equiv L_z/M = R^2 \dot{\phi}$$

to be the (constant) z component of the orbital angular momentum *per unit mass* of the star, we have that

$$\dot{\phi} = \frac{J_z}{R^2}$$

and

$$R\dot{\phi}^2 = \frac{J_z^2}{R^3}.$$

¹⁵Recall previous discussions of Φ in Sections 18.1 and 21.1.

Substituting into Eq. (25.19) gives

$$\ddot{R} = -\frac{\partial \Phi}{\partial R} + \frac{J_z^2}{R^3}.$$

To simplify the expression, we can define an *effective gravitational potential* that incorporates a term representing the portion of the star's kinetic energy per unit mass that is associated with its azimuthal motion ($v_\phi = R\dot{\phi} = J_z/R$):

$$\Phi_{\text{eff}}(R, z) \equiv \Phi(R, z) + \frac{J_z^2}{2R^2}. \quad (25.22)$$

Now the radial and vertical equations of motion can be written in the forms

$$\ddot{R} = -\frac{\partial \Phi_{\text{eff}}}{\partial R}, \quad (25.23)$$

$$\ddot{z} = -\frac{\partial \Phi_{\text{eff}}}{\partial z}, \quad (25.24)$$

the later expression coming from Eq. (25.21) with the azimuthal term in Φ_{eff} being independent of z .

To solve Eqs. (25.23) and (25.24) for the motion of the star through the galaxy, we must first determine the behavior of Φ_{eff} . Specifically, it would be helpful to know where the minima of the effective gravitational potential are located, since the star should attempt to settle into an orbit of minimum possible energy. In other words, we want to find the values of R and z for which

$$\frac{\partial \Phi_{\text{eff}}}{\partial R} = 0, \quad (25.25)$$

$$\frac{\partial \Phi_{\text{eff}}}{\partial z} = 0. \quad (25.26)$$

Given our assumption that the gravitational potential is symmetric about the midplane, it is clear that the second condition is satisfied at $z = 0$, because Φ , and therefore Φ_{eff} , must be either a local maximum or a local minimum there. Since Φ is always less than zero (cf., Eq. 21.1) and is assumed to be identically zero at infinity, it is necessary that Φ_{eff} be a minimum at $z = 0$.

The physical significance of the minimum in Φ_{eff} with respect to R can be uncovered by realizing that since J_z is a constant for the star's motion,

$$\frac{\partial \Phi_{\text{eff}}}{\partial R} = \frac{\partial \Phi}{\partial R} - \frac{J_z^2}{R^3} = 0$$

for some radius R_m in the galaxy's midplane ($z = 0$). This gives

$$\left. \frac{\partial \Phi}{\partial R} \right|_{(R_m, 0)} = \frac{J_z^2(R_m, 0)}{R_m^3}. \quad (25.27)$$

But since $J_z = Rv_\phi$, the right-hand side of the last expression is just

$$\frac{J_z^2}{R_m^3} = \frac{v_\phi^2}{R} \bigg|_{R_m},$$

the centripetal acceleration for perfectly circular motion. Furthermore, the left-hand side of Eq. (25.27) is simply the radial component of the gradient of the true gravitational potential, which is nothing more than the negative of the R component of the force per unit mass exerted on the star. Therefore, the last expression is the familiar equation of perfectly circular motion,

$$F_R(R_m) = - \frac{Mv_\phi^2}{R} \bigg|_{R_m},$$

and the minimum value for Φ_{eff} occurs when the star is executing perfectly circular orbital motion in the midplane of the spiral galaxy.

This is precisely the assumption we made in the last chapter regarding the motion of the local standard of rest at the position of the Sun. However, in Section 24.3 we also discovered that the Sun itself does not actually execute perfectly circular motion, but it does exhibit a peculiar motion with respect to the LSR, known as its solar motion. As a result, to learn more about the motion of our Sun (or, more generally, about the motions of stars in the plane of any spiral galaxy), it is necessary to explore deviations from the minimum value of Φ_{eff} .

To understand these first-order effects, we will expand Φ_{eff} about its minimum position $(R_m, 0)$ by means of a two-dimensional Taylor series. (This is similar to the procedure outlined in Problem 14.9.) Letting

$$\rho \equiv R - R_m$$

and using the subscript m to indicate that the leading constant and partial derivatives are being evaluated at the minimum position, we obtain

$$\begin{aligned} \Phi_{\text{eff}}(R, z) = & \Phi_{\text{eff},m} + \frac{\partial \Phi_{\text{eff}}}{\partial R} \bigg|_m \rho + \frac{\partial \Phi_{\text{eff}}}{\partial z} \bigg|_m z + \frac{1}{2} \frac{\partial^2 \Phi_{\text{eff}}}{\partial R \partial z} \bigg|_m \rho z \\ & + \frac{1}{2} \frac{\partial^2 \Phi_{\text{eff}}}{\partial R^2} \bigg|_m \rho^2 + \frac{1}{2} \frac{\partial^2 \Phi_{\text{eff}}}{\partial z^2} \bigg|_m z^2 + \dots \end{aligned} \quad (25.28)$$

The first term on the right-hand side is the constant minimum value in Φ_{eff} , and the two first-derivative terms are identically zero, since they were used to identify the point about which the expansion is being carried out (see Eqs. 25.25 and 25.26). The mixed-partial-derivative term is also identically zero because of the symmetry of Φ_{eff} about the $z = 0$

plane. Defining the constants

$$\kappa^2 \equiv \left. \frac{\partial^2 \Phi_{\text{eff}}}{\partial R^2} \right|_m, \quad (25.29)$$

$$\nu^2 \equiv \left. \frac{\partial^2 \Phi_{\text{eff}}}{\partial z^2} \right|_m, \quad (25.30)$$

and neglecting the remaining higher-order terms, the effective gravitational potential becomes

$$\Phi_{\text{eff}}(R, z) \simeq \Phi_{\text{eff},m} + \frac{1}{2} \kappa^2 \rho^2 + \frac{1}{2} \nu^2 z^2. \quad (25.31)$$

Finally, noting that $\ddot{\rho} = \ddot{R}$ and recalling Eqs. (25.23) and (25.24), we arrive at two of the three first-order expansions for the equations of motion about a perfectly circular orbit in the galaxy's midplane:

$$\ddot{\rho} \simeq -\kappa^2 \rho, \quad (25.32)$$

$$\ddot{z} \simeq -\nu^2 z, \quad (25.33)$$

which are the familiar differential equations of simple harmonic motion. Physically, Eqs. (25.32) and (25.33) represent the components of a star's acceleration relative to a point that is executing perfectly circular motion.

Equation (25.32) can be solved to give

$$\rho(t) = R(t) - R_m = A_R \sin \kappa t, \quad (25.34)$$

where κ is called the **epicycle frequency**.¹⁶ R_m is the radius of the energy-minimum circular orbit, and A_R is the amplitude of the radial oscillation. We have arbitrarily assumed that the star is passing through equilibrium and moving outward at some time $t = 0$.

Equation (25.33) is identical to the result obtained in Problem 24.27 for the sinusoidal oscillations of the Sun above and below the Galactic midplane, with ν being the **vertical oscillation frequency**. The star's position along the z -axis is given by

$$z(t) = A_z \sin(\nu t + \zeta), \quad (25.35)$$

where A_z is the amplitude of the oscillation in the z direction and ζ is a general phase shift between $\rho(t)$ and $z(t)$.

To help visualize the relationship between Φ_{eff} , as given in Eq. (25.31) and the equations of motion (Eqs. 25.34 and 25.35), imagine that the star is located inside a gravitational well described by a surface in the three-dimensional “space” of $(R, z, \Phi_{\text{eff}})$, depicted in Fig. 25.24. (The shape is similar to that of a bed sheet held at its corners, with $(R = R_m, z = 0)$ located at the lowest point.) Now consider the special case where the star is constrained to remain in the midplane ($A_z = 0$) but can oscillate about the path

¹⁶The choice of the term *epicycle frequency* will be described shortly.

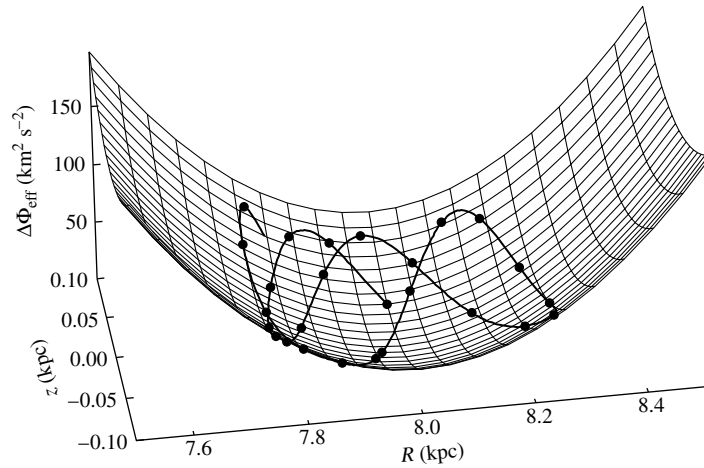


FIGURE 25.24 The effective gravitational potential well for a star that is executing a general first-order simple harmonic oscillation about a perfectly circular orbit near the midplane of a disk galaxy. In this case, the star is assumed to be oscillating about the equilibrium position ($R_m = 8$ kpc, $z = 0$). $\Delta\Phi_{\text{eff}} \equiv \Phi_{\text{eff}} - \Phi_{\text{eff},m}$.

of perfectly circular motion. This implies that the star is able to move only along the R -axis in Fig. 25.24. When the star is displaced from the minimum value of Φ_{eff} , a restoring force (given per unit mass by Eq. 25.32) will try to bring the star back toward equilibrium. Since the force is always directed toward the position of minimum Φ_{eff} , the star will accelerate toward the bottom of the potential well, overshoot the equilibrium position, and climb back up the opposite side. The star will then turn around and fall back down toward the bottom of the well again, continually executing simple harmonic motion. The same type of behavior would occur if the star were orbiting in a perfect circle projected onto the midplane ($A_R = 0$), but with superimposed vertical oscillations. However, since in general $\kappa \neq \nu$ and $\zeta \neq 0$, the general motion within the well can be quite complicated, as Fig. 25.24 illustrates.

We now have two expressions (Eqs. 25.34 and 25.35) that describe the motion of a star about an equilibrium position ($R = R_m$, $z = 0$) that is moving in a circular orbit. To complete our description of the approximate motion of the star, consider its azimuthal orbital angular speed, given by

$$\dot{\phi} = \frac{v_\phi}{R(t)} = \frac{J_z}{[R(t)]^2}.$$

But $R(t) = R_m + \rho(t) = R_m(1 + \rho(t)/R_m)$. Assuming that $\rho(t) \ll R_m$, as is required of our approximations, and using the binomial expansion theorem to first order,¹⁷

$$\dot{\phi} \approx \frac{J_z}{R_m^2} \left(1 - 2 \frac{\rho(t)}{R_m} \right).$$

¹⁷For $\delta \ll 1$, $(1 + \delta)^n \approx 1 + n\delta$.

Substituting the expression for $\rho(t)$ from Eq. (25.34) and integrating with respect to time, we find that

$$\phi(t) = \phi_0 + \frac{J_z}{R_m^2}t + \frac{2J_z}{\kappa R_m^3}A_R \cos \kappa t = \phi_0 + \Omega t + \frac{2\Omega}{\kappa R_m}A_R \cos \kappa t,$$

where $\Omega \equiv J_z/R_m^2$. The first two terms in this expression correspond to the perfectly circular orbit traced out by the equilibrium point, moving at a constant angular speed, Ω . The last term represents the oscillation of the star about the equilibrium point in the ϕ direction.

Finally, defining

$$\chi(t) \equiv [\phi(t) - (\phi_0 + \Omega t)] R_m$$

to be the difference in azimuthal position between the star and the equilibrium point, we have

$$\chi(t) = \frac{2\Omega}{\kappa} A_R \cos \kappa t. \quad (25.36)$$

The three equations—(25.34), (25.35), and (25.36)—represent the motions of the star in the R , z , and ϕ coordinates, respectively, about an equilibrium position that is moving in a perfect circle around the center of the galaxy and in the galactic midplane.

κ takes its name from an epicycle model for the first-order perturbation, as shown in Fig. 25.25. In general, in an inertial reference frame the star's orbit is not closed, but produces

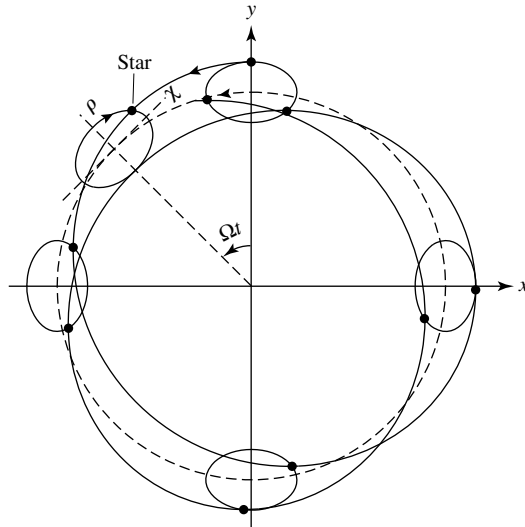


FIGURE 25.25 In an inertial reference frame a star's orbital motion in the galactic midplane (solid line) forms a nonclosing rosette pattern. In the first-order approximation, the motion can be imagined as being the combination of a retrograde orbit about an epicycle and the prograde orbit of the center of the epicycle about a perfect circle (dashed line). The dimensions of the epicycle have been exaggerated by a factor of five to illustrate the effect.

a rosette pattern. However, the star can be imagined as being located on an epicycle, with the center of the epicycle corresponding to the equilibrium position. As the star moves in a retrograde direction about the epicycle, it is carried alternately closer to and then farther from the galactic center. The epicycle is also oval in shape and has an axial ratio that is given by the ratios of the amplitudes of the oscillations in χ and ρ , or $2\Omega/\kappa$. The (χ, ρ) coordinate system of the epicycle rotates about the galaxy's center with the angular speed Ω (the angular speed of the equilibrium point) and is indicated in Fig. 25.25.¹⁸

Example 25.3.1. As we saw in Section 24.3, our Sun is moving relative to the perfectly circular motion of the LSR, as reflected in the Sun's peculiar solar motion. Since $u_\odot \neq 0$ (the R component of its peculiar motion), the Sun must have a nonzero epicycle frequency. Information about the radial component of the Sun's peculiar motion is also contained in the Oort constants, Eqs. (24.39) and (24.40), since they involve derivatives of the Sun's orbital speed, Θ_0 , with respect to Galactocentric radius, R_0 .

From the condition for perfectly circular motion, Eq. (25.27) gives

$$\frac{\partial \Phi}{\partial R} = \frac{J_z^2}{R^3} = \frac{\Theta_0^2}{R_0},$$

where the last expression arises because $J_z = R_0\Theta_0$ for the orbital angular momentum per unit mass at the solar Galactocentric distance. Substituting into the expression for the square of the epicycle frequency (Eq. 25.29) and making use of Eq. (25.22), we find that the solar epicycle frequency is

$$\kappa_0^2 = 2 \frac{\Theta_0}{R_0} \left[\frac{\Theta_0}{R_0} + \left. \frac{\partial \Theta_0}{\partial R} \right|_{R_0} \right]. \quad (25.37)$$

Rewriting this in terms of the Oort constants produces

$$\kappa_0^2 = -4B(A - B). \quad (25.38)$$

Using the values for the Oort constants obtained in Eqs. (24.47) and (24.48), $A = 14.8 \text{ km s}^{-1} \text{ kpc}^{-1}$ and $B = -12.4 \text{ km s}^{-1} \text{ kpc}^{-1}$, respectively, the epicycle frequency for our Sun is $\kappa_0 = 36.7 \text{ km s}^{-1} \text{ kpc}^{-1} = 1.2 \times 10^{-15} \text{ rad s}^{-1}$. (Note that when kiloparsecs are converted to kilometers, κ_0 has units of rad s^{-1} .) This value for the epicycle frequency corresponds to an oscillation period of $P = 2\pi/\kappa_0 = 170 \text{ Myr}$.

The ratio of the Sun's epicycle frequency to its orbital angular speed (or orbital frequency, $\Omega_0 = \Theta_0/R_0 = A - B$; Eq. 24.43) is

$$\frac{\kappa_0}{\Omega_0} = 2 \left(\frac{-B}{A - B} \right)^{1/2} = 1.35.$$

Therefore, the Sun executes 1.35 epicycle oscillations for every orbit around the Galactic center.

¹⁸This model bears a strong resemblance to the epicycle–deferent models of planetary motion devised by Hipparchus and Ptolemy, illustrated in Fig. 1.3. However, in those ancient planetary models, the epicycles were assumed to be perfectly circular.

The axial ratio of the Sun's epicycle is given by

$$\frac{\chi_{\max}}{\rho_{\max}} = \frac{2\Omega_0}{\kappa_0} \simeq 1.5.$$

Closed Orbits in Noninertial Frames

The number of oscillations per orbit about the galaxy's center is equal to the ratio of the star's epicycle frequency to its orbital angular speed. If the ratio κ/Ω is a ratio of integers, the orbit is closed. Like the Sun's, however, most stellar orbits are not closed and a rosette pattern results. But in a noninertial reference frame that is rotating with a *local* angular pattern speed, $\Omega_{\ell p} = \Omega$, relative to the inertial frame, the star's path would appear to be very simple, tracing out a *closed orbit* that is retrograde and centered at a distance R_m from the galaxy's center (see Fig. 25.25). Such a reference frame corresponds to the epicycle's own coordinate system in which the equilibrium point is stationary and the closed path simply traces the epicycle itself.

To obtain a closed orbit in a noninertial reference frame, we need not necessarily choose an angular pattern speed that equals the unperturbed orbital angular speed Ω . Instead, we could choose to have the star complete n orbits as seen in the rotating frame while executing m epicycle oscillations (where n and m are positive or negative integers), after which time the star would be back at its starting point. That is, we could choose $m(\Omega - \Omega_{\ell p}) = n\kappa$, or

$$\Omega_{\ell p}(R) = \Omega(R) - \frac{n}{m}\kappa(R). \quad (25.39)$$

Note that this is a local pattern speed, so $\Omega_{\ell p}$ is a function of R . Although in principle, there are an infinite number of local pattern speeds at each R , only a small number of values for n and m produce substantial enhancements in mass density.¹⁹

Figure 25.26 shows the rotation of a coordinate system for which the local pattern speed corresponds to $(n = 1, m = 2)$, together with the position of the star at one point. Figure 25.27 shows the motion of the star as seen in four rotating reference frames with $(n, m) = (0, 1), (1, 2), (2, 3)$, and $(1, 4)$. The position of the star that was indicated in Fig. 25.26 corresponds to the same position in Fig. 25.27(a).

Now imagine a large number of stars at various distances R from the center of a spiral galaxy, all observed in a reference frame rotating with the *global* angular pattern speed Ω_{gp} . If we consider the case $(n, m) = (1, 2)$ and if the local pattern speed $\Omega_{\ell p} = \Omega(R) - \kappa(R)/2$ is a constant for all values of R , then we can set $\Omega_{gp} = \Omega_{\ell p}$. Seen from the noninertial frame, the resulting orbital patterns could be nested with their major axes aligned, as illustrated in Fig. 25.28(a). The structure that results bears a significant resemblance to the bars present in roughly two-thirds of all spiral galaxies. Of course, we could also orient each successive oval-shaped orbit so that its major axis is rotated slightly relative to the one immediately interior to it. In this example the result is a trailing two-armed grand-design spiral wave

¹⁹This problem is analogous to situations encountered within the Solar System. For example, the orbital resonances of Saturn's moons (primarily Mimas) with particles in the planet's ring system produce gaps in the rings. In addition, small integer ratios between Jupiter's orbital period and the orbital periods of asteroids result in either increases or decreases in the number of asteroids having certain orbital radii.

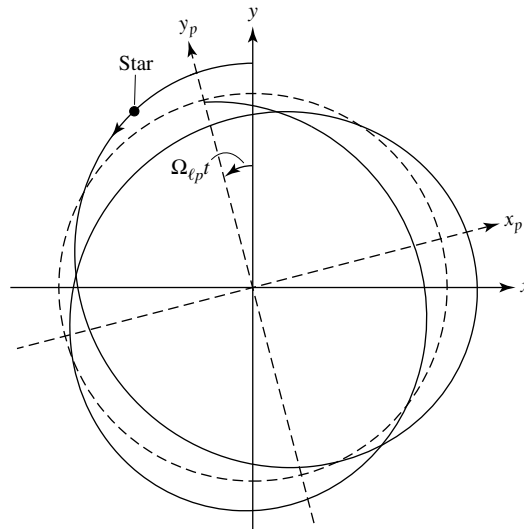


FIGURE 25.26 A noninertial coordinate system, (x_p, y_p) , rotating within a fixed inertial frame (x, y) with the local angular pattern speed $\Omega_{\ell p} = \Omega - \kappa/2$, for which $(n = 1, m = 2)$. The dashed line corresponds to the perfectly circular motion of the equilibrium point, and the solid line represents the orbital motion of the star in the galaxy's inertial reference frame. The position of the star on the diagram, along with the position of the noninertial coordinate system, corresponds to one-eighth of the orbital period of the equilibrium point.

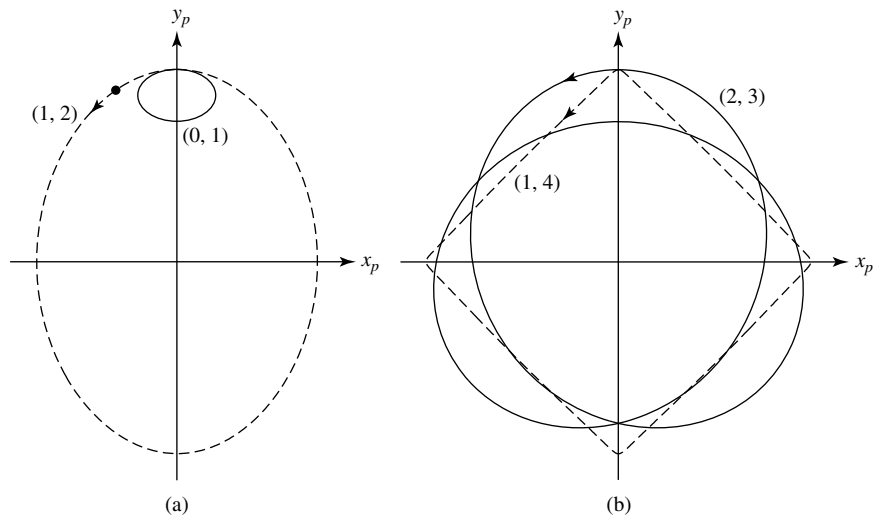


FIGURE 25.27 Stars appear to trace out closed orbits in noninertial coordinate systems that rotate with appropriately chosen angular pattern speeds. The motion of a star as it would appear in coordinate systems having local angular pattern speeds corresponding to (n, m) values of (a) $(0, 1)$ = solid line, $(1, 2)$ = dashed line, (b) $(2, 3)$ = solid line, and $(1, 4)$ = dashed line.

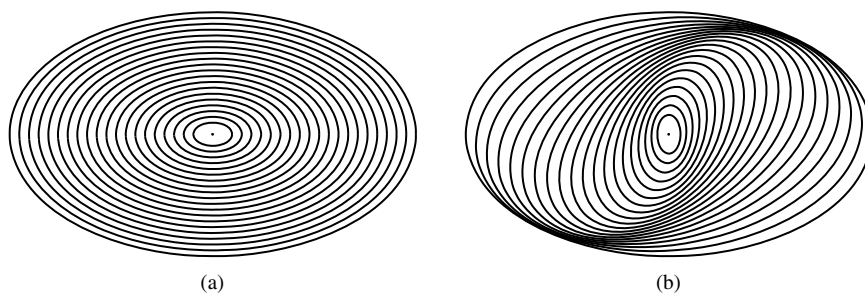


FIGURE 25.28 (a) Nested oval orbits with aligned major axes, as seen in a reference frame rotating with the global angular pattern speed ($n = 1, m = 2$), or $\Omega_{gp} = \Omega - \kappa/2$. The result is a bar-like structure. (b) Each oval is rotated relative to the orbit immediately interior to it. The result is a two-armed grand-design spiral density wave.

pattern, shown in Fig. 25.28(b). Twisting the ovals in the opposite sense would result in a leading two-armed spiral. The two-armed spiral M51 (Fig. 25.17) is an example of a trailing-arm, ($n = 1, m = 2$) pattern structure, while the four-armed spiral M101 [Fig. 25.5(c)] is an ($n = 1, m = 4$) system. Patterns with $m = 2$ are the most common type of density wave structure.

It is important to remember that the individual stars are following their own orbits in an inertial reference frame and that these orbits appear only as simplified oval shapes in a reference frame that is rotating with the local angular pattern speed. Furthermore, even in the rotating frame the stars themselves are still moving along the oval orbits. Only the spiral pattern appears to be static in that frame (if we still assume that $\Omega_{gp} = \Omega_{\ell p}$, independent of R); in the nonrotating inertial frame the spiral pattern will appear to move with an angular speed of Ω_{gp} . It is the “traffic jam” of stars becoming packed together where their oval orbits approach one another that leads to the density waves.

The stability of the structures shown in Fig. 25.28 depends crucially on whether $\Omega_{\ell p} = \Omega(R) - \kappa(R)/2$ is actually independent of R —that is, on whether there is an appropriate global value of Ω_{gp} . Figure 25.29 shows curves of $\Omega(R) - n\kappa(R)/m$ with several ratios of n/m for one model of our Galaxy. Notice that $\Omega - \kappa/2$ is nearly flat over a wide range of values for R , a fact first realized by the Swedish astronomer Bertil Lindblad (1895–1965). The same general behavior is exhibited by a large number of spiral galaxies and probably accounts for the prevalence of two-armed spirals. Of course, $\Omega - \kappa/2$ is not exactly constant with respect to galactocentric radius, so some drifting of epicycle orbits relative to one another does occur, leading once again to a winding problem. However, in this case the winding occurs for density waves rather than material waves, and because the relative drift is slower, the winding takes about five times longer to develop. If a means can be found to stabilize the galaxy against this remnant winding effect, the original Lin–Shu hypothesis of a quasistatic density wave will be realized.

Lindblad, Corotation, and Ultraharmonic Resonances

Since $\Omega - \kappa/2$ is not exactly constant across the entire galactic disk, stars at varying radii do not encounter the spiral arms at precisely the same point in their epicyclic paths on

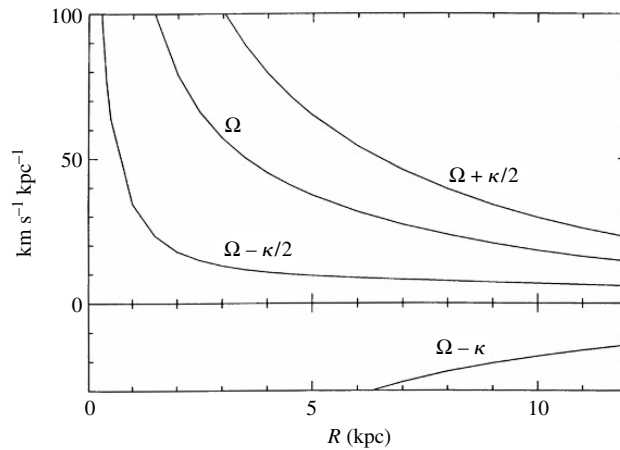


FIGURE 25.29 The Bahcall–Soneira model of our Galaxy has been used to construct the functions $\Omega_{\ell p} = \Omega - n\kappa/m$ for various ratios of n/m . Note the nearly constant curve for $n/m = 1/2$ over much of the Galactic disk. (Figure adapted from Binney and Tremaine, *Galactic Dynamics*, Princeton University Press, Princeton, NJ, 1987.)

successive passages through the density waves. However, there are certain radii for which this can happen and a resonance develops, analogous to the resonance that can occur when a spring is forced to oscillate at its natural frequency. For instance, if a star is at its maximum value of χ (Eq. 25.36) each time it encounters a density wave, the perturbation it experiences due to the local increase in density and gravitational potential will always be in the same sense, and the effect will be cumulative. A similar argument applies to each of the local pattern speeds, such as $\Omega_{\ell p} = \Omega(R)$ and $\Omega_{\ell p} = \Omega(R) + \kappa/2$.²⁰

This amplification can occur at several radii in a galaxy, depending on its mass distribution and its resulting rotation curve. An **inner Lindblad resonance** exists when the local angular pattern speed of the star equals the global angular pattern speed of the density wave for the case when $\Omega_{gp} = \Omega - \kappa/2$ for $n/m = 1/2$. It is possible that either zero, one, or two inner Lindblad radii may exist for a given galaxy, depending on the shape of the rotation curve. A **corotation resonance** can occur if $\Omega_{gp} = \Omega$ for some value of R , and an **outer Lindblad resonance** might exist if $\Omega_{gp} = \Omega + \kappa/2$. The **ultraharmonic resonance** may also develop for $\Omega_{gp} = \Omega - \kappa/4$. The inner rings of galaxies such as NGC 7096 and M81 [Figs. 25.5(a) and (b)] are apparently due to either inner Lindblad resonances or ultraharmonic resonances, while the outer rings of galaxies like NGC 4340 [Fig. 25.4(f)] are found at the outer Lindblad resonance. The positions of the Lindblad and corotation resonances are depicted in Fig. 25.30 for two different rotation curves.

Resonances have the effect that epicycle oscillation amplitudes increase dramatically. As a result, collisions of gas clouds should also increase significantly at resonance positions, and energy will be dissipated. Consequently, resonance locations (if they exist in a specific galaxy) can actually lead to damping of spiral waves unless other processes are capable of

²⁰Note that we are now going beyond the assumption made on page 969 that the density waves make an insignificant contribution to the gravitational field.